



part I

enlightenment and the anti- bayesian reaction

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causes in the air

Sometime during the 1740s, the Reverend Thomas Bayes made the ingenious discovery that bears his name but then mysteriously abandoned it. It was rediscovered independently by a different and far more renowned man, Pierre Simon Laplace, who gave it its modern mathematical form and scientific application—and then moved on to other methods. Although Bayes' rule drew the attention of the greatest statisticians of the twentieth century, some of them vilified both the method and its adherents, crushed it, and declared it dead. Yet at the same time, it solved practical questions that were unanswerable by any other means: the defenders of Captain Dreyfus used it to demonstrate his innocence; insurance actuaries used it to set rates; Alan Turing used it to decode the German Enigma cipher and arguably save the Allies from losing the Second World War; the U.S. Navy used it to search for a missing H-bomb and to locate Soviet subs; RAND Corporation used it to assess the likelihood of a nuclear accident; and Harvard and Chicago researchers used it to verify the authorship of the Federalist Papers. In discovering its value for science, many supporters underwent a near-religious conversion yet had to conceal their use of Bayes' rule and pretend they employed something else. It was not until the twenty-first century that the method lost its stigma and was widely and enthusiastically embraced. The story began with a simple thought experiment.

Because Bayes' gravestone says he died in 1761 at the age of 59, we know he lived during England's struggle to recover from nearly two centuries of religious strife, civil war, and regicide. As a member of the Presbyterian Church,

a religious denomination persecuted for refusing to support the Church of England, he was considered a Dissenter or Non-Conformist. During his grandfather's generation, 2,000 Dissenters died in English prisons. By Bayes' time, mathematics was split along religious and political lines, and many productive mathematicians were amateurs because, as Dissenters, they were barred from English universities.¹

Unable to earn a degree in England, Bayes studied theology and presumably mathematics at the University of Edinburgh in Presbyterian Scotland, where, happily for him, academic standards were much more rigorous. In 1711 he left for London, where his clergyman father ordained him and apparently employed him as an assistant minister.

Persecution turned many English Dissenters into feisty critics, and in his late 20s Bayes took a stand on a hot theological issue: can the presence of evil be reconciled with God's presumed beneficence? In 1731 he wrote a pamphlet—a kind of blog—declaring that God gives people “the greatest happiness of which they are capable.”

During his 40s, Bayes' interests in mathematics and theology began to tightly intertwine. An Irish-Anglican bishop—George Berkeley, for whom the University of California's flagship campus is named—published an inflammatory pamphlet attacking Dissenting mathematicians, calculus, abstract mathematics, the revered Isaac Newton, and all other “free-thinkers” and “infidel mathematicians” who believed that reason could illuminate any subject. Berkeley's pamphlet was the most spectacular event in British mathematics during the 1700s.

Leaping into the pamphlet wars again, Bayes published a piece defending and explaining Newton's calculus. This was his only mathematical publication during his lifetime. Shortly thereafter, in 1742, five men, including a close friend of Newton's, nominated Bayes for membership in the Royal Society. His nomination avoided any hint of controversy and described him as “a Gentleman of known merit, well skilled in Geometry and all parts of Mathematical and Philosophical Learning.” The Royal Society was not the professional organization it is today; it was a private organization of dues-paying amateurs from the landed gentry. But it played a vital role because amateurs would produce some of the era's breakthroughs.

About this time Bayes joined a second group of up-to-date amateur mathematicians. He had moved to a small congregation in a fashionable resort, the cold-water spa Tunbridge Wells. As an independently wealthy bachelor—his family had made a fortune manufacturing Sheffield steel

cutlery—he rented rooms, apparently from a Dissenting family. His religious duties—one Sunday sermon a week—were light. And spa etiquette permitted Dissenters, Jews, Roman Catholics, and even foreigners to mix with English society, even with wealthy earls, as they could not elsewhere.

A frequent visitor to Tunbridge Wells, Philip, the Second Earl of Stanhope, had been passionately interested in mathematics since childhood, but his guardian had banned its study as insufficiently genteel. When Stanhope was 20 and free to do as he liked, he seldom raised his eyes from Euclid. According to the bluestocking Elizabeth Montagu, Stanhope was “always making mathematical scratches in his pocket-book, so that one half the people took him for a conjuror, and the other half for a fool.” Because of either his aristocratic position or his late start Stanhope never published anything of his own. Instead, he became England’s foremost patron of mathematicians.

The earl and the Royal Society’s energetic secretary, John Canton, operated an informal network of peer reviewers who critiqued one another’s work. At some point Bayes joined the network. One day, for example, Stanhope sent Bayes a copy of a draft paper by a mathematician named Patrick Murdoch. Bayes disagreed with some of it and sent his comments back to Stanhope, who forwarded them to Murdoch, who in turn replied through Stanhope, and so on around and around. The relationship between the young earl and the older Reverend Bayes seems to have ripened into friendship, however, because Stanhope paid Bayes at least one personal visit at Tunbridge Wells, saved two bundles of his mathematical papers in the Stanhope estate’s library, and even subscribed to his series of sermons.

Another incendiary mix of religion and mathematics exploded over England in 1748, when the Scottish philosopher David Hume published an essay attacking some of Christianity’s fundamental narratives. Hume believed that we can’t be absolutely certain about anything that is based only on traditional beliefs, testimony, habitual relationships, or cause and effect. In short, we can rely only on what we learn from experience.

Because God was regarded as the First Cause of everything, Hume’s skepticism about cause-and-effect relationships was especially unsettling. Hume argued that certain objects are constantly associated with each other. But the fact that umbrellas and rain appear together does not mean that umbrellas cause rain. The fact that the sun has risen thousands of times does not guarantee that it will do so the next day. And, most important, the “design of the world” does not prove the existence of a creator, an ultimate cause. Because we can seldom be certain that a particular cause will have a

particular effect, we must be content with finding only probable causes and probable effects. In criticizing concepts about cause and effect Hume was undermining Christianity's core beliefs.

Hume's essay was nonmathematical, but it had profound scientific implications. Many mathematicians and scientists believed fervently that natural laws did indeed prove the existence of God, their First Cause. As an eminent mathematician Abraham de Moivre wrote in his influential book *Doctrine of Chances*, calculations about natural events would eventually reveal the underlying order of the universe and its exquisite "Wisdom and Design."

With Hume's doubts about cause and effect swirling about, Bayes began to consider ways to treat the issue mathematically. Today, probability, the mathematics of uncertainty, would be the obvious tool, but during the early 1700s probability barely existed. Its only extensive application was to gambling, where it dealt with such basic issues as the odds of getting four aces in one poker hand. De Moivre, who had spent several years in French prisons because he was a Protestant, had already solved that problem by working from cause to effect. But no one had figured out how to turn his work around backward to ask the so-called inverse question from effect to cause: what if a poker player deals himself four aces in each of three consecutive hands? What is the underlying chance (or cause) that his deck is loaded?

We don't know precisely what piqued Bayes' interest in the inverse probability problem. He had read de Moivre's book, and the Earl of Stanhope was interested in probability as it applied to gambling. Alternatively, it could have been the broad issues raised by Newton's theory of gravitation. Newton, who had died 20 years before, had stressed the importance of relying on observations, developed his theory of gravitation to explain them, and then used his theory to predict new observations. But Newton had not explained the cause of gravity or wrestled with the problem of how true his theory might be. Finally, Bayes' interest may have been stimulated by Hume's philosophical essay. In any event, problems involving cause and effect and uncertainty filled the air, and Bayes set out to deal with them quantitatively.

Crystallizing the essence of the inverse probability problem in his mind, Bayes decided that his goal was to learn the approximate probability of a future event he knew nothing about except its past, that is, the number of times it had occurred or failed to occur. To quantify the problem, he needed a number, and sometime between 1746 and 1749 he hit on an ingenious solution. As a starting point he would simply invent a number—he called it a guess—and refine it later as he gathered more information.

Next, he devised a thought experiment, a 1700s version of a computer simulation. Stripping the problem to its basics, Bayes imagined a square table so level that a ball thrown on it would have the same chance of landing on one spot as on any other. Subsequent generations would call his construction a billiard table, but as a Dissenting minister Bayes would have disapproved of such games, and his experiment did not involve balls bouncing off table edges or colliding with one another. As he envisioned it, a ball rolled randomly on the table could stop with equal probability anywhere.

We can imagine him sitting with his back to the table so he cannot see anything on it. On a piece of paper he draws a square to represent the surface of the table. He begins by having an associate toss an imaginary cue ball onto the pretend tabletop. Because his back is turned, Bayes does not know where the cue ball has landed.

Next, we picture him asking his colleague to throw a second ball onto the table and report whether it landed to the right or left of the cue ball. If to the left, Bayes realizes that the cue ball is more likely to sit toward the right side of the table. Again Bayes' friend throws the ball and reports only whether it lands to the right or left of the cue ball. If to the right, Bayes realizes that the cue can't be on the far right-hand edge of the table.

He asks his colleague to make throw after throw after throw; gamblers and mathematicians already knew that the more times they tossed a coin, the more trustworthy their conclusions would be. What Bayes discovered is that, as more and more balls were thrown, each new piece of information made his imaginary cue ball wobble back and forth within a more limited area.

As an extreme case, if all the subsequent tosses fell to the right of the first ball, Bayes would have to conclude that it probably sat on the far left-hand margin of his table. By contrast, if all the tosses landed to the left of the first ball, it probably sat on the far right. Eventually, given enough tosses of the ball, Bayes could narrow the range of places where the cue ball was apt to be.

Bayes' genius was to take the idea of narrowing down the range of positions for the cue ball and—based on this meager information—infer that it had landed somewhere between two bounds. This approach could not produce a right answer. Bayes could never know precisely where the cue ball landed, but he could tell with increasing confidence that it was most probably within a particular range. Bayes' simple, limited system thus moved from observations about the world back to their probable origin or cause. Using his knowledge of the present (the left and right positions of the tossed balls), Bayes had figured out how to say something about the past

(the position of the first ball). He could even judge how confident he could be about his conclusion.

Conceptually, Bayes' system was simple. We modify our opinions with objective information: Initial Beliefs (our guess where the cue ball landed) + Recent Objective Data (whether the most recent ball landed to the left or right of our original guess) = A New and Improved Belief. Eventually, names were assigned to each part of his method: Prior for the probability of the initial belief; Likelihood for the probability of other hypotheses with objective new data; and Posterior for the probability of the newly revised belief. Each time the system is recalculated, the posterior becomes the prior of the new iteration. It was an evolving system, which each new bit of information pushed closer and closer to certitude. In short:

Prior times likelihood is proportional to the posterior.

(In the more technical language of the statistician, the likelihood is the probability of competing hypotheses for the fixed data that have been observed. However, Andrew Dale, a South African historian of statistics, simplified the matter considerably when he observed, "Put somewhat rudely, the likelihood is what remains of Bayes's Theorem once the prior is removed from the discussion.")²

As a special case about balls thrown randomly onto a flat table, Bayes' rule is uncontroversial. But Bayes wanted to cover *every* case involving uncertainty, even cases where nothing whatsoever was known about their history—in his words, where we "absolutely know nothing antecedently to any trials."³ This expansion of his table experiment to cover any uncertain situation would trigger 150 years of misunderstanding and bitter attacks.

Two especially popular targets for attack were Bayes' guesswork and his suggested shortcut.

First, Bayes *guessed* the likely value of his initial belief (the cue ball's position, later known as the prior). In his own words, he decided to make "a guess whereabouts it's [sic] probability is, and . . . [then] see the chance that the guess is right." Future critics would be horrified at the idea of using a mere hunch—a subjective belief—in objective and rigorous mathematics.

Even worse, Bayes added that if he *did not know enough* to distinguish the position of the balls on his table, he would assume they were equally likely to fall anywhere on it. Assuming equal probabilities was a pragmatic approach for dealing with uncertain circumstances. The practice was rooted in traditional Christianity and the Roman Catholic Church's ban on usury. In uncertain situ-

ations such as annuities or marine insurance policies, all parties were assigned equal shares and divided profits equally. Even prominent mathematicians assigned equal probabilities to gambling odds by assuming, with a remarkable lack of realism, that all tennis players or fighting cocks were equally skillful.

In time, the practice of assigning equal probabilities acquired a number of names, including equal priors, equal *a priori*'s, equiprobability, uniform distribution probability, and the law of insufficient reason (meaning that without enough data to assign specific probabilities, equal ones would suffice). Despite their venerable history, equal probabilities would become a lightning rod for complaints that Bayes was quantifying ignorance.

Today, some historians try to absolve him by saying he may have applied equal probabilities to his data (the subsequent throws) rather than to the initial, so-called prior toss. But this is also guesswork. And for many working statisticians, the question is irrelevant because in the tightly circumscribed case of balls that can roll anywhere on a carefully leveled surface both produce the same mathematical results.

Whatever Bayes meant, the damage was done. For years to come, the message seemed clear: priors be damned. At this point, Bayes ended his discussion.

He may have mentioned his discovery to others. In 1749 someone told a physician named David Hartley something that sounds suspiciously like Bayes' rule. Hartley was a Royal Society member who believed in cause-and-effect relationships. In 1749 he wrote that "an ingenious friend has communicated to me a Solution of the inverse problem . . . which shews that we may hope to determine the Proportions, and by degrees, the whole Nature, of unknown Causes, by a sufficient Observation of their Effects." Who was this ingenious friend? Modern-day sleuths have suggested Bayes or Stanhope, and in 1999 Stephen M. Stigler of the University of Chicago suggested that Nicholas Saunderson, a blind Cambridge mathematician, made the discovery instead of Bayes. No matter who talked about it, it seems highly unlikely that anyone other than Bayes made the breakthrough. Hartley used terminology that is almost identical to Bayes' published essay, and no one who read the article between its publication in 1764 and 1999 doubted Bayes' authorship. If there had been any question about the author's identity, it is hard to imagine Bayes' editor or his publisher not saying something publicly. Thirty years later Price was still referring to the work as that of Thomas Bayes.

Although Bayes' idea was discussed in Royal Society circles, he himself seems not to have believed in it. Instead of sending it off to the Royal Society for publication, he buried it among his papers, where it sat for roughly a

decade. Only because he filed it between memoranda dated 1746 and 1749 can we conclude that he achieved his breakthrough sometime during the late 1740s, perhaps shortly after the publication of Hume's essay in 1748.

Bayes' reason for suppressing his essay can hardly have been fear of controversy; he had plunged twice into Britain's pamphlet wars. Perhaps he thought his discovery was useless; but if a pious clergyman like Bayes thought his work could prove the existence of God, surely he would have published it. Some thought Bayes was too modest. Others wondered whether he was unsure about his mathematics. Whatever the reason, Bayes made an important contribution to a significant problem—and suppressed it. It was the first of several times that “Bayes' rule” would spring to life only to disappear again from view.

Bayes' discovery was still gathering dust when he died in 1761. At that point relatives asked Bayes' young friend Richard Price to examine Bayes' mathematical papers.

Price, another Presbyterian minister and amateur mathematician, achieved fame later as an advocate of civil liberties and of the American and French revolutions. His admirers included the Continental Congress, which asked him to emigrate and manage its finances; Benjamin Franklin, who nominated him for the Royal Society; Thomas Jefferson, who asked him to write to Virginia's youths about the evils of slavery; John Adams and the feminist Mary Wollstonecraft, who attended his church; the prison reformer John Howard, who was his best friend; and Joseph Priestley, the discoverer of oxygen, who said, “I question whether Dr. Price ever had a superior.” When Yale University conferred two honorary degrees in 1781, it gave one to George Washington and the other to Price. An English magazine thought Price would go down in American history beside Franklin, Washington, Lafayette, and Paine. Yet today Price is known primarily for the help he gave his friend Bayes.

Sorting through Bayes' papers, Price found “an imperfect solution of one of the most difficult problems in the doctrine of chances.” It was Bayes' essay on the probability of causes, on moving from observations about the real world back to their most probable cause.

At first Price saw no reason to devote much time to the essay. Mathematical infelicities and imperfections marred the manuscript, and it looked impractical. Its continual iterations—throwing the ball over and over again and recalculating the formula each time—produced large numbers that would be difficult to calculate.

But once Price decided Bayes' essay was the answer to Hume's attack

on causation, he began preparing it for publication. Devoting “a good deal of labour” to it on and off for almost two years, he added missing references and citations and deleted extraneous background details in Bayes’ derivations. Lamentably, he also threw out his friend’s introduction, so we’ll never know precisely how much the edited essay reflects Bayes’ own thinking.

In a cover letter to the Royal Society, Price supplied a religious reason for publishing the essay. In moving mathematically from observations of the natural world inversely back to its ultimate cause, the theorem aimed to show that “the world must be the effect of the wisdom and power of an intelligent cause; and thus to confirm . . . from final causes . . . the existence of the Deity.” Bayes himself was more reticent; his part of the essay does not mention God.

A year later the Royal Society’s *Philosophical Transactions* published “An Essay toward solving a Problem in the Doctrine of Chances.” The title avoided religious controversy by highlighting the method’s gambling applications. Critiquing Hume a few years later, Price used Bayes for the first and only time. As far as we know, no one else mentioned the essay for the next 17 years, when Price would again bring Bayes’ rule to light.

By modern standards, we should refer to the Bayes-Price rule. Price discovered Bayes’ work, recognized its importance, corrected it, contributed to the article, and found a use for it. The modern convention of employing Bayes’ name alone is unfair but so entrenched that anything else makes little sense.

Although it was ignored for years, Bayes’ solution to the inverse probability of causes was a masterpiece. He transformed probability from a gambler’s measure of frequency into a measure of informed belief. A card player could start by believing his opponent played with a straight deck and then modify his opinion each time a new hand was dealt. Eventually, the gambler could wind up with a better assessment of his opponent’s honesty.

Bayes combined judgments based on prior hunches with probabilities based on repeatable experiments. He introduced the signature features of Bayesian methods: an initial belief modified by objective new information. He could move from observations of the world to abstractions about their probable cause. And he discovered the long-sought grail of probability, what future mathematicians would call the probability of causes, the principle of inverse probability, Bayesian statistics, or simply Bayes’ rule.

Given the revered status of his work today, it is also important to recognize what Bayes did not do. He did not produce the modern version of Bayes’

rule. He did not even employ an algebraic equation; he used Newton's old-fashioned geometric notation to calculate and add areas. Nor did he develop his theorem into a powerful mathematical method. Above all, unlike Price, he did not mention Hume, religion, or God.

Instead, he cautiously confined himself to the probability of events and did not mention hypothesizing, predicting, deciding, or taking action. He did not suggest possible uses for his work, whether in theology, science, or social science. Future generations would extend Bayes' discovery to do all these things and to solve a myriad of practical problems. Bayes did not even name his breakthrough. It would be called the probability of causes or inverse probability for the next 200 years. It would not be named Bayesian until the 1950s.

In short, Bayes took the first steps. He composed the prelude for what was to come.

For the next two centuries few read the Bayes-Price article. In the end, this is the story of two friends, Dissenting clergymen and amateur mathematicians, whose labor had almost no impact. Almost, that is, except on the one person capable of doing something about it, the great French mathematician Pierre Simon Laplace.

the man who did everything

Just across the English Channel from Tunbridge Wells, about the time that Thomas Bayes was imagining his perfectly smooth table, the mayor of a tiny village in Normandy was celebrating the birth of a son, Pierre Simon Laplace, the future Einstein of his age.

Pierre Simon, born on March 23, 1749, and baptized two days later, came from several generations of literate and respected dignitaries. His mother's relatives were well-to-do farmers, but she died when he was young, and he never referred to her. His father kept the stagecoach inn in picturesque Beaumont-en-Auge, was a leader of the community's 472 inhabitants, and served 30 years as mayor. By the time Pierre Simon was a teenager his father seems to have been his only close relative. In years to come Pierre Simon's decision to become a mathematician would shatter their relationship almost irretrievably.¹

Fortunately for the boy there was never any question about his getting an education. Attending school was becoming the norm in France in the 1700s, an enormous revolution fueled by the Catholic Church's fight against Protestant heresy and by parents convinced that education would enrich their children spiritually, intellectually, and financially. The question was, what kind of schooling?

Decades of religious warfare between Protestants and Catholics and several horrendous famines caused by cold weather had made France a determinedly secular country intent on developing its resources. Pierre Simon could have studied modern science and geometry in one of the country's many new secular schools. Instead, the elder Laplace enrolled his son in a local primary and secondary school where Benedictine monks produced clergy

for the church and soldiers, lawyers, and bureaucrats for the crown. Thanks to the patronage of the Duke of Orleans, local day students like Pierre Simon attended free. The curriculum was conservative and Latin-based, heavy on copying, memorization, and philosophy. But it left Laplace with a fabulous memory and almost unbelievable perseverance.

Although the monks probably did not know it, they were competing with the French Enlightenment for the child's attention. Contemporaries called it the Century of Lights and the Age of Science and Reason, and the popularization of science was its most important intellectual phenomenon. Given the almost dizzying curiosity of the times, it is not surprising that, shortly after his tenth birthday, Pierre Simon was profoundly affected by a spectacular scientific prediction.²

Decades before, the English astronomer Edmond Halley had predicted the reappearance of the long-tailed comet that now bears his name. A trio of French astronomers, Alexis Claude Clairaut, Joseph Lalande, and Nicole-Reine Lepaute, the wife of a celebrated clockmaker, solved a difficult three-body problem and discovered that the gravitational pull of Jupiter and Saturn would delay the arrival of Halley's comet. The French astronomers accurately pinpointed the date—mid-April 1759 plus or minus a month—when Europeans would be able to see the comet returning from its orbit around the sun. The comet's appearance on schedule and on course electrified Europeans. Years later Laplace said it was the event that made his generation realize that extraordinary events like comets, eclipses, and severe droughts were caused not by divine anger but by natural laws that mathematics could reveal.

Laplace's extraordinary mathematical ability may not yet have been apparent when he turned 17 in 1766, because he did not go to the University of Paris, which had a strong science faculty. Instead he went to the University of Caen, which was closer to home and had a solid theological program suitable for a future cleric.

Yet even Caen had mathematical firebrands offering advanced lectures on differential and integral calculus. While English mathematicians were getting mired in Newton's awkward geometric version of calculus, their rivals on the Continent were using Gottfried Leibniz's more supple algebraic calculus. With it, they were forming equations and discovering a fabulous wealth of enticing new information about planets, their masses and details of their orbits. Laplace emerged from Caen a swashbuckling mathematical virtuoso eager to take on the scientific world. He had also become, no doubt to his father's horror, a religious skeptic.

At graduation Laplace faced an anguishing dilemma. His master's degree permitted him to take either the priestly vows of celibacy or the title of abbé, signifying a low-ranking clergyman who could marry and inherit property. Abbés did not have good reputations; Voltaire called them "that indefinable being which is neither ecclesiastic nor secular . . . young men, who are known for their debauchery."³ An engraving of the period, "What Does the Abbé Think of It?" shows the clergyman peering appreciatively down a lady's bosom as she dresses.⁴ Still, the elder Laplace wanted his son to become a clergyman.

If Laplace had been willing to become an abbé, his father might have helped him financially, and Laplace could have combined church and science. A number of abbés supported themselves in science, the most famous being Jean Antoine Nollet, who demonstrated spectacular physics experiments to the paying public. For the edification of the king and queen of France, Nollet sent a charge of static electricity through a line of 180 soldiers to make them leap comically into the air. Two abbés were even elected to the prestigious Royal Academy of Sciences. Still, the lot of most abbé-scientists was neither lucrative nor intellectually challenging. The majority found low-level jobs tutoring the sons of rich nobles or teaching elementary mathematics and science in secondary schools. University-level opportunities were limited because during the 1700s professors transmitted knowledge from the past instead of doing original research.

But Caen had convinced Laplace that he wanted to do something quite new. He wanted to be a full-time, professional, secular, mathematical researcher. And he wanted to explore the new algebra-generated, data-rich world of science. To his father, an ambitious man in bucolic France, a career in mathematics must have seemed preposterous.

Young Laplace made his move in the summer of 1769, shortly after completing his studies at Caen. He left Normandy and traveled to Paris, clutching a letter of recommendation to Jean Le Rond d'Alembert, the most powerful mathematician of the age, one of Europe's most notorious anticlerics, and the object of almost incessant Jesuit attacks. D'Alembert was a star of the Enlightenment and the chief spokesman for the *Encyclopédie*, which was making an enormous body of empirical knowledge universally available, scientific, and free of religious dogma. By throwing in his lot with d'Alembert, Laplace effectively cut his ties to the Catholic Church. We can only imagine his father's reaction, but we know that Laplace did not return home for 20 years and did not attend the old man's funeral.

Once in Paris, Laplace immediately approached the great d'Alembert and showed him a four-page student essay on inertia. Years later Laplace could still recite passages from it. Although besieged by applicants, d'Alembert was so impressed that within days he had arranged a paying job for Laplace as an instructor of mathematics at the new secular, mathematics-based Royal Military School for the younger sons of minor nobles. The school, located behind Les Invalides in Paris, provided Laplace with a salary, housing, meals, and money for wood to heat his room in winter. It was precisely the kind of job he had hoped to avoid.

Laplace could have tried to find work applying mathematics to practical problems in one of the monarchy's numerous research establishments or manufacturing plants. Many mathematically talented young men from modest families were employed in such institutions. But Laplace and his mentor were aiming far higher. Laplace wanted the challenge of doing basic research full time. And to do that, as d'Alembert must have told him, he had to get elected to the Royal Academy of Sciences.

In striking contrast to the amateurism of the Royal Society of London, the French Royal Academy of Sciences was the most professional scientific institution in Europe. Although aristocratic amateurs could become honorary members, the organization's highest ranks were composed of working scientists chosen by merit and paid to observe, collect, and investigate facts free of dogma; to publish their findings after peer review; and to advise the government on technical issues like patents. To augment their low salaries, academicians could use their prestige to cobble together various part-time jobs.

Without financial support from the church or his father, however, Laplace had to work fast. Since most academy members were chosen on the basis of a long record of solid accomplishment, he would have to be elected over the heads of more senior men. And for that to happen, he needed to make a spectacular impact.

D'Alembert, who had made Newton's revolution the focus of French mathematics, urged Laplace to concentrate on astronomy. D'Alembert had a clear problem in mind.

Over the previous two centuries mathematical astronomy had made great strides. Nicolaus Copernicus had moved Earth from the center of the solar system to a modest but accurate position among the planets; Johannes Kepler had connected the celestial bodies by simple laws; and Newton had introduced the concept of gravity. But Newton had described the motions

of heavenly bodies roughly and without explanation. His death in 1727 left Laplace's generation an enormous challenge: showing that gravitation was not a hypothesis but a fundamental law of nature.

Astronomy was the era's most quantified and respected science, and only it could test Newton's theories by explaining precisely how gravitation affects the movements of tides, interacting planets and comets, our moon, and the shape of Earth and other planets. Forty years of empirical data had been collected, but, as d'Alembert warned, a single exception could bring the entire edifice tumbling down.

The burning scientific question of the day was whether the universe was stable. If Newton's gravitational force operates throughout the universe, why don't the planets collide with each other and cause the cosmic Armageddon described in the biblical book of Revelation? Was the end of the world at hand?

Astronomers had long been aware of alarming evidence suggesting that the solar system was inherently unstable. Comparing the actual positions of the most remote known planets with centuries-old astronomical observations, they could see that Jupiter was slowly accelerating in its orbit around the sun while Saturn was slowing down. Eventually, they thought, Jupiter would smash into the sun, and Saturn would spin off into space. The problem of predicting the motions of many interacting bodies over long periods of time is complex even today, and Newton concluded that God's miraculous intervention kept the heavens in equilibrium. Responding to the challenge, Laplace decided to make the stability of the universe his lifework. He said his tool would be mathematics and it would be like a telescope in the hands of an astronomer.

For a short time Laplace actually considered modifying Newton's theory by making gravity vary with a body's velocity as well as with its mass and distance. He also wondered fleetingly whether comets might be disturbing the orbits of Jupiter and Saturn. But he changed his mind almost immediately. The problem was not Newton's theory. The problem was the data astronomers used.

Newton's system of gravitation could be accepted as true only if it agreed with precise measurements, but observational astronomy was awash with information, some of it uncertain and inadequate. Working on the problem of Jupiter and Saturn, for example, Laplace would use observations made by Chinese astronomers in 1100 BC, Chaldeans in 600 BC, Greeks in 200 BC, Romans in AD 100, and Arabs in AD 1000. Obviously, not all data were

equally valuable. How to resolve errors, known delicately as discrepancies, was anybody's guess.

The French academy was tackling the problem by encouraging the development of more precise telescopes and graduated arcs. And as algebra improved instrumentation, experimentalists were producing more quantitative results. In a veritable information explosion, the sheer collection and systemization of data accelerated through the Western world. Just as the number of known plant and animal species expanded enormously during the 1700s, so did knowledge about the physical universe. Even as Laplace arrived in Paris, the French and British academies were sending trained observers with state-of-the-art instrumentation to 120 carefully selected locations around the globe to time Venus crossing the face of the sun; this was a critical part of Capt. James Cook's original mission to the South Seas. By comparing all the measurements, French mathematicians would determine the approximate distance between the sun and Earth, a fundamental natural constant that would tell them the size of the solar system. But sometimes even up-to-date expeditions provided contradictory data about whether, for instance, Earth was shaped like an American football or a pumpkin.

Dealing with large amounts of complex data was emerging as a major scientific problem. Given a wealth of observations, how could scientists evaluate the facts at their disposal and choose the most valid? Observational astronomers typically averaged their three best observations of a particular phenomenon, but the practice was as straightforward as it was ad hoc; no one had ever tried to prove its validity empirically or theoretically. The mathematical theory of errors was in its infancy.

Problems were ripe for the picking and, with his eye on membership in the Royal Academy, Laplace bombarded the society with 13 papers in five years. He submitted hundreds of pages of powerful and original mathematics needed in astronomy, celestial mechanics, and important related issues. Astutely, he timed his reports to appear when openings occurred in the academy's membership. The secretary of the academy, the Marquis de Condorcet, wrote that never before had the society seen "anyone so young, present to it in so little time, so many important *Mémoires*, and on such diverse and such difficult matters."⁵

Academy members considered Laplace for membership six times but rejected him repeatedly in favor of more senior scientists. D'Alembert complained furiously that the organization refused to recognize talent. Laplace considered emigrating to Prussia or Russia to work in their academies.

During this frustrating period Laplace spent his free afternoons digging in the mathematical literature in the Royal Military School's 4,000-volume library. Analyzing large amounts of data was a formidable problem, and Laplace was already beginning to think it would require a fundamentally new way of thinking. He was beginning to see probability as a way to deal with the uncertainties pervading many events and their causes. Browsing in the library's stacks, he discovered an old book on gambling probability, *The Doctrine of Chances*, by Abraham de Moivre. The book had appeared in three editions between 1718 and 1756, and Laplace may have read the 1756 version. Thomas Bayes had studied an earlier edition.

Reading de Moivre, Laplace became more and more convinced that probability might help him deal with uncertainties in the solar system. Probability barely existed as a mathematical term, much less as a theory. Outside of gambling, it was applied in rudimentary form to philosophical questions like the existence of God and to commercial risk, including contracts, marine and life insurance, annuities, and money lending.

Laplace's growing interest in probability created a diplomatic problem of some delicacy because d'Alembert believed probability was too subjective for science. Young as he was, Laplace was confident enough in his mathematical judgment to disagree with his powerful patron. To Laplace, the movements of celestial bodies seemed so complex that he could not hope for precise solutions. Probability would not give him absolute answers, but it might show him which data were more likely to be correct. He began thinking about a method for deducing the probable causes of divergent, error-filled observations in astronomy. He was feeling his way toward a broad general theory for moving mathematically from known events back to their most probable causes. Continental mathematicians did not know yet about Bayes' discovery, so Laplace called his idea "the probability of causes" and "the probability of causes and future events, derived from past events."⁶

Wrestling with the mathematics of probability in 1773, he reflected on its philosophical counterpoint. In a paper submitted and read to the academy in March, the former abbé compared ignorant mankind, not with God but with an imaginary intelligence capable of knowing All. Because humans can never know everything with certainty, probability is the mathematical expression of our ignorance: "We owe to the frailty of the human mind one of the most delicate and ingenious of mathematical theories, namely the science of chance or probabilities."⁷

The essay was a grand combination of mathematics, metaphysics, and

the heavens that Laplace held to his entire life. His search for a probability of causes and his view of the deity were deeply congenial. Laplace was all of one piece and for that reason all the more formidable. He often said he did not believe in God, and not even his biographer could decide whether he was an atheist or a deist. But his probability of causes was a mathematical expression of the universe, and for the rest of his days he updated his theories about God and the probability of causes as new evidence became available.

Laplace was struggling with probability when one day, ten years after the publication of Bayes' essay, he picked up an astronomy journal and was shocked to read that others might be hot on the same trail. They were not, but the threat of competition galvanized him. Dusting off one of his discarded manuscripts, Laplace transformed it into a broad method for determining the most likely causes of events and phenomena. He called it "Mémoire on the Probability of the Causes Given Events."

It provided the first version of what today we call Bayes' rule, Bayesian probability, or Bayesian statistical inference. Not yet recognizable as the modern Bayes' rule, it was a one-step process for moving backward, or inversely, from an effect to its most likely cause. As a mathematician in a gambling-addicted culture, Laplace knew how to work out the gambler's future odds of an event knowing its cause (the dice). But he wanted to solve scientific problems, and in real life he did not always know the gambler's odds and often had doubts about what numbers to put into his calculations. In a giant and intellectually nimble leap, he realized he could inject these uncertainties into his thinking by considering *all* possible causes and then choosing among them.

Laplace did not state his idea as an equation. He intuited it as a principle and described it only in words: the probability of a cause (given an event) is proportional to the probability of the event (given its cause). Laplace did not translate his theory into algebra at this point, but modern readers might find it helpful to see what his statement would look like today:

$$P(C|E) = \frac{P(E|C)}{\sum P(E|C')}$$

where $P(C|E)$ is the probability of a particular cause (given the data), and $P(E|C)$ represents the probability of an event or datum (given that cause). The sign in the denominator represented with Newton's sigma sign makes the total probability of all possible causes add up to one.

Armed with his principle, Laplace could do everything Thomas Bayes could have done—as long as he accepted the restrictive assumption that all his possible causes or hypotheses were equally likely. Laplace's goal, however, was far more ambitious. As a scientist, he needed to study the various possible causes of a phenomenon and then determine the best one. He did not yet know how to do that mathematically. He would need to make two more major breakthroughs and spend decades in thought.

Laplace's principle, the proportionality between probable events and their probable causes, seems simple today. But he was the first mathematician to work with large data sets, and the proportionality of cause and effect would make it feasible to make complex numerical calculations using only goose quills and ink pots.

In a *mémoire* read aloud to the academy, Laplace first applied his new probability of causes to two gambling problems. In each case he understood intuitively what should happen but got bogged down trying to prove it mathematically. First, he imagined an urn filled with an unknown ratio of black and white tickets (his cause). He drew a number of tickets from the urn and, based on that experience, asked for the probability that his next ticket would be white. Then in a frustrating battle to prove the answer he wrote no fewer than 45 equations covering four quarto-sized pages.

His second gambling problem involved piquet, a game requiring both luck and skill. Two people start playing but stop midway through the game and have to figure out how to divide the kitty by estimating their relative skill levels (the cause). Again, Laplace understood instinctively how to solve the problem but could not yet do so mathematically.

After dealing with gambling, which he loathed, Laplace moved happily on to the critical scientific problem faced by working astronomers. How should they deal with different observations of the same phenomenon? Three of the era's biggest scientific problems involved gravitational attraction on the motions of our moon, the motions of the planets Jupiter and Saturn, and the shape of the Earth. Even if observers repeated their measurements at the same time and place with the same instrument, their results could be slightly different each time. Trying to calculate a midvalue for such discrepant observations, Laplace limited himself to three observations but still needed seven pages of equations to formulate the problem. Scientifically, he understood the right answer—average the three data points—but he would have no mathematical justification for doing so until 1810, when, without using the probability of causes, he invented the central limit theorem.

Although Bayes originated the probability of causes, Laplace clearly discovered his version on his own. Laplace was 15 when the Bayes-Price essay was published; it appeared in an English-language journal for the English gentry and was apparently never mentioned again. Even French scientists who kept up with foreign journals thought Laplace was first and congratulated him wholeheartedly on his originality.

Mathematics confirms that Laplace discovered the principle independently. Bayes solved a special problem about a flat table using a two-step process that involved a prior guess and new data. Laplace did not yet know about the initial guess but dealt with the problem generally, making it useful for a variety of problems. Bayes laboriously explained and illustrated why uniform probabilities were permissible; Laplace assumed them instinctively. The Englishman wanted to know the range of probabilities that something will happen in light of previous experience. Laplace wanted more: as a working scientist, he wanted to know the probability that certain measurements and numerical values associated with a phenomenon were realistic. If Bayes and Price searched for the probability that, on the basis of today's puddles, it had rained yesterday and would rain tomorrow, Laplace asked for the probability that a particular amount of rain would fall and then refined his opinion over and over with new information to get a better value. Laplace's method was immensely influential; scientists did not pay Bayes serious heed until the twentieth century.

Most strikingly of all, Laplace at 25 was already steadfastly determined to develop his new method and make it useful. For the next 40 years he would work to clarify, simplify, expand, generalize, prove, and apply his new rule. Yet while Laplace became the indisputable intellectual giant of Bayes' rule, it represented only a small portion of his career. He also made important advances in celestial mechanics, mathematics, physics, biology, Earth science, and statistics. He juggled projects, moving from one to another and then back to the first. Happily blazing trails through every field of science known to his age, he transformed and mathematized everything he touched. He never stopped being thrilled by examples of Newton's theory.

Although he was fast becoming the leading scientist of his era, the academy waited five years before electing him a member on March 31, 1773. A few weeks later he was formally inducted into the world's leading scientific organization. His *mémoire* on the probability of causes was published a year later, in 1774. At the age of 24, Laplace was a professional researcher. The academy's annual stipend, together with his teaching salary, would help

support him while he refined his research on celestial mechanics and the probability of causes.

Laplace was still grappling with probability in 1781, when Richard Price visited Paris and told Condorcet about Bayes' discovery. Laplace immediately latched onto the Englishman's ingenious invention, the starting guess, and incorporated it into his own, earlier version of the probability of causes. Strictly speaking, he did not produce a new formula but rather a statement about the first formula assuming equal probabilities for the causes. The statement gave him confidence that he was on the right track and told him that as long as all his prior hypotheses were equally probable, his earlier principle of 1774 was correct.⁸

Laplace could now confidently marry his intuitive grasp of a scientific situation with the eighteenth century's passion for new and precise scientific discoveries. Every time he got new information he could use the answer from his last solution as the starting point for another calculation. And by assuming that all his initial hypotheses were equally probable he could even derive his theorem.

As Academy secretary, Condorcet wrote an introduction to Laplace's essay and explained Bayes' contribution. Laplace later publicly credited Bayes with being first when he wrote, "The theory whose principles I explained some years after, . . . he accomplished in an acute and very ingenious, though slightly awkward, manner."⁹

Over the next decade, however, Laplace would realize with increasing clarity and frustration that his mathematics had shortcomings. It limited him to assigning equal probabilities to each of his initial hypotheses. As a scientist, he disapproved. If his method was ever going to reflect the actual state of affairs, he needed to be able to differentiate dubious data from more valid observations. Calling all events or observations equally probable could be true only theoretically. Many dice, for example, that appeared perfectly cubed were actually skewed. In one case he started by assigning players equal probabilities of winning, but with each round of play their respective skills emerged and their probabilities changed. "The science of chances must be used with care and must be modified when we pass from the mathematical case to the physical," he counseled.¹⁰

Moreover, as a pragmatist, he realized he had to confront a serious technical difficulty. Probability problems require multiplying numbers over and over, whether tossing coin after coin or measuring and remeasuring

an observation. The process generated huge numbers—nothing as large as those common today but definitely cumbersome for a man working alone without mechanical or electronic aids. (He did not even get an assistant to help with calculations until about 1785.)

Laplace was never one to shrink from difficult computations, but, as he complained, probability problems were often impossible because they presented great difficulties and numbers raised to “very high powers.”¹¹ He could use logarithms and an early generating function that he considered inadequate. But to illustrate how tedious calculations with big numbers could be, he described multiplying $20,000 \times 19,999 \times 19,998 \times 19,997$, etc. and then dividing by $1 \times 2 \times 3 \times 4$ up to 10,000. In another case he bet in a lottery only to realize he could not calculate its formula numerically; the French monarchy’s winning number had 90 digits, drawn five at a time.

Such big-number problems were new. Newton had calculated with geometry, not numbers. Many mathematicians, like Bayes, used thought experiments to separate real problems from abstract and methodological issues. But Laplace wanted to use mathematics to illuminate natural phenomena, and he insisted that theories had to be based on actual fact. Probability was propelling him into an unmanageable world.

Armed with the Bayes–Price starting point, Laplace broke partway through the logjam that had stymied him for seven years. So far he had concentrated primarily on probability as a way to resolve error-prone astronomical observations. Now he switched gears to concentrate on finding the most probable causes of known events. To do so, he needed to practice with a big database of real and reliable values. But astronomy seldom provided extensive or controlled data, and the social sciences often involved so many possible causes that algebraic equations were useless.

Only one large amalgamation of truly trustworthy numbers existed in the 1700s: parish records of births, christenings, marriages, and deaths. In 1771 the French government ordered all provincial officials to report birth and death figures regularly to Paris; and three years later, the Royal Academy published 60 years of data for the Paris region. The figures confirmed what the Englishman John Graunt had discovered in 1662: slightly more boys than girls were born, in a ratio that remained constant over many years. Scientists had long assumed that the ratio, like other newly discovered regularities in nature, must be the result of “Divine Providence.” Laplace disagreed.

Soon he was assessing not gambling or astronomical statistics but infants. For anyone interested in large numbers, babies were ideal. First, they

came in binomials, either boys or girls, and eighteenth-century mathematicians already knew how to treat binomials. Second, infants arrived in abundance and, as Laplace emphasized, “It is necessary in this delicate research to employ sufficiently large numbers in view of the small difference that exists between . . . the births of boys and girls.”¹² When the great naturalist Comte de Buffon discovered a small village in Burgundy where, for five years running, more girls had been born than boys, he asked whether this village invalidated Laplace’s hypotheses. Absolutely not, Laplace replied firmly. A study based on a few facts cannot overrule a much larger one.

The calculations would be formidable. For example, if he had started with a 52:48 ratio of newborn boys to girls and a sample of 58,000 boys, Laplace would have had to multiply .52 by itself 57,999 times—and then do a similar calculation for girls. This was definitely not something anyone, not even the indomitable Laplace, wanted to do by hand.

He started out, however, as Bayes had suggested, by pragmatically assigning equal probabilities to all his initial hunches, whether 50–50, 33–33–33, or 25–25–25–25. Because their sums equal one, multiplication would be easier. He employed equal probabilities only provisionally, as a starting point, and his final hypotheses would depend on all the observational data he could add.

Next, he tried to confirm that Graunt was correct about the probability of a boy’s birth being larger than 50%. He was building the foundation of the modern theory of testing statistical hypotheses. Poring over records of christenings in Paris and births in London, he was soon willing to bet that boys would outnumber girls for the next 179 years in Paris and for the next 8,605 years in London. “It would be extraordinary if it was the effect of chance,” he wrote, tut-tutting that people really should make sure of their facts before theorizing about them.¹³

To transform probability’s large numbers into smaller, more manageable terms Laplace invented a multitude of mathematical shortcuts and clever approximations. Among them were new generating functions, transforms, and asymptotic expansions. Computers have made many of his shortcuts unnecessary, but generating functions remain deeply embedded in mathematical analyses used for practical applications. Laplace used generating functions as a form of mathematical wizardry to trick a function he could deal with into providing him with the function he really wanted.

To Laplace, these mathematical pyrotechnics seemed as obvious as common sense. To students’ frustration, he sprinkled his reports with phrases like, “It is easy to see, it is easy to extend, it is easy to apply, it is obvious

that. . . .”¹⁴ When a confused student once asked how he had jumped intuitively from one equation to another, Laplace had to work hard to reconstruct his thought process.

He was soon asking whether boys were more apt to be born in certain geographic regions. Perhaps “climate, food or customs . . . facilitates the birth of boys” in London.¹⁵ Over the next 30-odd years Laplace collected birth ratios from Naples in the south, St. Petersburg in the north, and French provinces in between. He concluded that climate could not explain the disparity in births. But would more boys than girls always be born? As each additional piece of evidence appeared, Laplace found his probabilities approaching certainty “at a dramatically increasing rate.”

He was refining hunches with objective data. In building a mathematical model of scientific thinking, where a reasonable person could develop a hypothesis and then evaluate it relentlessly in light of new knowledge, he became the first modern Bayesian. His system was enormously sensitive to new information. Just as each throw of a coin increases the probability of its being fair or rigged, so each additional birth record narrowed the range of uncertainties. Eventually, Laplace decided that the probability of boys exceeding girls was as “certain as any other moral truth” with an extremely tiny margin of being wrong.¹⁶

Generalizing from babies, he found a way to determine not just the probability of simple events, like the birth of one boy, but also the probability of future composite events like an entire year of births—even when the probability of simple events (whether the next newborn will be male) was uncertain. By 1786 he was determining the influence of past events on the probability of future events and wondering how big his sample of newborns had to be. By then Laplace saw probability as the primary way to overcome uncertainty. Pounding the point home in one short paragraph, he wrote, “Probability is relative in part to this *ignorance*, in part to our knowledge . . . a state of indecision, . . . it’s impossible to announce with certainty.”¹⁷

Persevering for years, he used insights gained in one science to shed light on others, researching a puzzle and inventing a mathematical technique to resolve it, integrating, approximating, and generalizing broadly when there was no other way to proceed. Like a modern researcher, he competed and collaborated with others and published reports on his interim progress as he went. Above all, he was tenacious. Twenty-five years later he was still eagerly testing his probability of causes with new information. He combed 65 years’ worth of orphanage registries, asked friends in Egypt and Alexander

von Humboldt in Central America about birth ratios there, and called on naturalists to check the animal kingdom. Finally, in 1812, after decades of work, he cautiously concluded that the birth of more boys than girls seemed to be “a general law for the human race.”¹⁸

To test his rule on a larger sample Laplace decided in 1781 to determine the size of the French population, the thermometer of its health and prosperity. A conscientious administrator in eastern France had carefully counted heads in several parishes; to estimate the population of the entire nation, he recommended multiplying the annual number of births in France by 26. His proposal produced what was thought to be France’s population, approximately 25.3 million. But no one knew how accurate his estimate was. Today’s demographers believe that France’s population had actually grown rapidly, to almost 28 million, because of fewer famines and because a government-trained midwife was touring the countryside promoting the use of soap and boiling water during childbirth.

Using his probability of causes, Laplace combined his prior information from parish records about births and deaths throughout France with his new information about headcounts in eastern France. He was adjusting estimates of the nation’s population with more precise information from particular regions. In 1786 he reached a figure closer to modern estimates and calculated odds of 1,000 to 1 that his estimate was off by less than half a million. In 1802 he was able to advise Napoleon Bonaparte that a new census should be augmented with detailed samples of about a million residents in 30 representative departments scattered equally around France.

As he worked on his birth and census studies during the monarchy’s last years, Laplace became involved in an inflammatory debate about France’s judicial system. Condorcet believed the social sciences should be as quantifiable as the physical sciences. To help transform absolutist France into an English-style constitutional monarchy, he wanted Laplace to use mathematics to explore a variety of issues. How confident can we be in a sentence handed down by judge or jury? How probable is it that voting by an assembly or judicial tribunal will establish the truth? Laplace agreed to apply his new theory of probability to questions about electoral procedures, the credibility of witnesses, decision making by judicial panels and juries, and procedures of representative bodies and judicial panels.

Laplace took a dim view of most court judgments in France. Forensic science did not exist, so judicial systems everywhere relied on witness testi-

mony. Taking a witness's statement for an event, Laplace asked the probability that the witness or the judge might be truthful, misled, or simply mistaken. He estimated the prior odds of an accused person's guilt at 50–50 and the probability that a juror was being truthful somewhat higher. Even at that, if a jury of eight voted by simple majority, the chance that they judged the accused's guilt wrong would be $65/256$, or more than one in four. Thus for both mathematical and religious reasons Laplace sided with the Enlightenment's most radical demand, the abolition of capital punishment: "The possibility of atoning for these errors is the strongest argument of philosophers who have wanted to abolish the death penalty."¹⁹ Laplace also used his rule for more complicated cases where a court must decide among contradictory witnesses or where the reliability of testimony decreases with each telling. For Laplace, these questions demonstrated that ancient biblical accounts by the Apostles lacked credibility.

While still counting babies, Laplace returned to study the seeming instability of Saturn and Jupiter's orbits, the problem that had helped sensitize him early in his career to uncertain data. He did not, however, use his new knowledge of probability to solve this important problem. He used other methods between 1785 and 1788 to determine that Jupiter and Saturn oscillate gently in an 877-year cycle around the sun and that the moon orbits Earth in a cycle millions of years long. The orbits of Jupiter, Saturn, and the moon were not exceptions to Newton's gravitation but thrilling examples of it. The solar system was in equilibrium, and the world would not end. This discovery was the biggest advance in physical astronomy since Newton's law of gravity.

Despite Laplace's astounding productivity, his life as a professional scientist was financially precarious. Fortunately, Paris in the 1700s had more educational institutions and scientific opportunities than anywhere else on Earth, and academy members could patch jobs together to make a respectable living. Laplace tripled his income by examining artillery and naval engineering students three or four months a year and serving as a scientist in the Duke of Orleans' entourage. His increasingly secure position also gave him access to the government statistics he needed to develop and test his probability of causes.

At the age of 39, with a bright future ahead of him, Laplace married 18-year-old Marie Anne Charlotte Courty de Romange. The average age of

marriage for French women was 27, but Marie Anne came from a prosperous and recently ennobled family with multiple ties to his financial and social circle. A small street off the Boulevard Saint-Germain is named Courty for her family. The Laplaces would have two children; contraception, whether coitus interruptus or pessaries, was common, and the church itself campaigned against multiple childbirths because they endangered the lives of mothers. Some 16 months after the wedding a Parisian mob stormed the Bastille, and the French Revolution began.

After the revolutionary government was attacked by foreign monarchies, France spent a decade at war. Few scientists or engineers emigrated, even during the Reign of Terror. Mobilized for the national defense, they organized the conscription of soldiers, collected raw materials for gunpowder, supervised munitions factories, drew military maps, and invented a secret weapon, reconnaissance balloons. Laplace worked throughout the upheaval and served as the central figure in one of the Revolution's most important scientific projects, the metric reform to standardize weights and measures. It was Laplace who named the meter, centimeter, and millimeter.

Nevertheless, during the 18 months of the Terror, as almost 17,000 French were executed and half a million imprisoned, his position became increasingly precarious. Radicals attacked the elite Royal Academy of Sciences, and publications denounced him as a modern charlatan and a "Newtonian idolator." A month after the Royal Academy was abolished Laplace was arrested on suspicion of disloyalty to the Revolution but neighbors interceded and he was released the next day at 4 a.m. A few months later he was purged from the metric system commission as not "worthy of confidence as to [his] republican virtues and [his] hatred of kings."²⁰ His assistant, Jean-Baptiste Delambre, was arrested while measuring the meridian for the meter and then released. At one point Laplace was relieved of his part-time job examining artillery students, only to be given the same job at the École Polytechnique. Seven scientists, including several of Laplace's closest friends and supporters, died during the Terror. Unlike Laplace, who took no part in radical politics, they had identified themselves with particular political factions. The most famous was Antoine Lavoisier, guillotined because he had been a royal tax collector. Condorcet, trying to escape from Paris, died in jail.

The Revolution, however, transformed science from a popular hobby into a full-fledged profession. Laplace emerged from the chaos as a dean of French science, charged with building new secular educational institutions and training the next generation of scientists. For almost 50 years—from the

1780s until his death in 1827—France led world science as no other country has before or since. And for 30 of those years Laplace was among the most influential scientists of all time.

As the best-selling author of books about the celestial system and the law of gravity, Laplace dedicated two volumes to a rising young general, Napoleon Bonaparte. Laplace had launched Napoleon on his military career by giving him a passing exam grade in military school. The two never became personal friends, but Napoleon appointed Laplace minister of the interior for a short time and then appointed him to the largely honorary Senate with a handsome salary and generous expense account that made him quite a rich man. Mme Laplace became a lady-in-waiting to Napoleon's sister and received her own salary. With additional financing from Napoleon, Laplace and his friend the chemist Claude Berthollet turned their country homes in Arceuil, outside Paris, into the world's only center for young postdoctoral scientists.

At a reception in Josephine Bonaparte's rose garden at Malmaison in 1802, the emperor, who was trying to engineer a rapprochement with the papacy, started a celebrated argument with Laplace about God, astronomy, and the heavens.

"And who is the author of all this?" Napoleon demanded.

Laplace replied calmly that a chain of natural causes would account for the construction and preservation of the celestial system.

Napoleon complained that "Newton spoke of God in his book. I have perused yours but failed to find His name even once. Why?"

"Sire," Laplace replied magisterially, "I have no need of that hypothesis."²¹

Laplace's answer, so different from Price's idea that Bayes' rule could prove the existence of God, became a symbol of a centuries-long process that would eventually exclude religion from the scientific study of physical phenomena. Laplace had long since separated his probability of causes from religious considerations: "The true object of the physical sciences is not the search for primary causes [that is, God] but the search for laws according to which phenomena are produced."²² Scientific explanations of natural phenomena were triumphs of civilization whereas theological debates were fruitless because they could never be resolved.

Laplace continued his research throughout France's political upheavals. In 1810 he announced the central limit theorem, one of the great scientific and statistical discoveries of all time. It asserts that, with some exceptions, any average of a large number of similar terms will have a normal, bell-shaped distribution. Suddenly, the easy-to-use bell curve was a real mathemati-

cal construct. Laplace's probability of causes had limited him to binomial problems, but his final proof of the central limit theorem let him deal with almost any kind of data.

In providing the mathematical justification for taking the mean of many data points, the central limit theorem had a profound effect on the future of Bayes' rule. At the age of 62, Laplace, its chief creator and proponent, made a remarkable about-face. He switched allegiances to an alternate, frequency-based approach he had also developed. From 1811 until his death 16 years later Laplace relied primarily on this approach, which twentieth-century theoreticians would use to almost obliterate Bayes' rule.

Laplace made the change because he realized that where large amounts of data were concerned, both approaches generally produce much the same results. The probability of causes was still useful in particularly uncertain cases because it was more powerful than frequentism. But science matured during Laplace's lifetime. By the 1800s mathematicians had much more reliable data than they had had in his youth and dealing with trustworthy data was easier with frequentism. Mathematicians did not learn until the mid-twentieth century that, even with great amounts of data, the two methods can sometimes seriously disagree.

Looking back in 1813 on his 40-year quest to develop the probability of causes, Laplace described it as the primary method for researching unknown or complicated causes of natural phenomena. He referred to it fondly as his source of large numbers and the inspiration behind his development and use of generating functions.

And finally, in the climax of one small part of his career, he proved the elegant, general version of his theorem that we now call Bayes' rule. He had intuited its principle as a young man in 1774. In 1781 he found a way to use Bayes' two-step process to derive the formula by making certain restrictive assumptions. Between 1810 and 1814 he finally realized what the general theorem had to be. It was the formula he had been dreaming about, one broad enough to allow him to distinguish highly probable hypotheses from less valid ones. With it, the entire process of learning from evidence was displayed:

$$P(C|E) = \frac{P(E|C) P_{\text{prior}}(C)}{\sum P(E|C') P_{\text{prior}}(C')}$$

In modern terms, the equation says that $P(C|E)$, the probability of a hypothesis (given information), equals $P_{\text{prior}}(C)$, our initial estimate of its probability, times $P(E|C)$, the probability of each new piece of information

(under the hypothesis), divided by the sum of the probabilities of the data in all possible hypotheses.

Undergraduates today study Laplace's first version of the equation, which deals with discrete events such as coin tosses and births. Advanced and graduate students and researchers use calculus with his later equation to work with observations on a continuous range between two values, for example, all the temperatures between 32 and 33 degrees. With it, Laplace could estimate a value as being within such and such a range with a particular degree of probability.

Laplace had owned Bayes' rule in all but name since 1781. The formula, the method, and its masterful utilization all belong to Pierre Simon Laplace. He made probability-based statistics commonplace. By transforming a theory of gambling into practical mathematics, Laplace's work dominated probability and statistics for a century. "In my mind," Glenn Shafer of Rutgers University observed, "Laplace did everything, and we just read stuff back into Thomas Bayes. Laplace put it into modern terms. In a sense, everything is Laplacean."²³

If advancing the world's knowledge is important, Bayes' rule should be called Laplace's rule or, in modern parlance, BPL for Bayes-Price-Laplace. Sadly, a half century of usage forces us to give Bayes' name to what was really Laplace's achievement.

Since discovering his first version of Bayes' rule in 1774, Laplace had used it primarily to develop new mathematical techniques and had applied it most extensively to the social sciences, that is, demography and judicial reform. Not until 1815, at the age of 66, did he apply it to his first love, astronomy. He had received some astonishingly accurate tables compiled by his assistant Alexis Bouvard, the director of the Paris Observatory. Using Laplace's probability of causes, Bouvard had calculated a large number of observations about the masses of Jupiter and Saturn, estimated the possible error for each entry, and then predicted the probable masses of the planets. Laplace was so delighted with the tables that, despite his aversion to gambling, he used Bayes' rule to place a famous bet with his readers: odds were 11,000 to 1 that Bouvard's results for Saturn were off by less than 1%. For Jupiter, the odds were a million to one. Space-age technology confirms that Laplace and Bouvard should have won both bets.

Late in his career, Laplace also applied his probability of causes to a variety of calculations in Earth science, notably to the tides and to changes in barometric pressure. He used a nonnumerical common-sense version of his probability of causes to advance his famous nebular hypothesis: that the

planets and their satellites in our solar system originated in a swirl of dust. And he compared three hypotheses about the orbits of 100 comets to confirm what he already knew: that the comets most probably originate within the sun's sphere of influence.

After the fall of Napoleon, France's new king, Louis XVIII, bestowed the hereditary title of marquis on Laplace, the son of a village innkeeper. And on March 5, 1827, at the age of 78, Laplace died, almost exactly 100 years after his idol, Isaac Newton.

Eulogies hailed Laplace as the Newton of France. He had brought modern science to students, governments, and the reading public and had developed probability into a formidable method for handling unknown and complex causes of natural phenomena. And in one small, relatively insignificant portion of his lifework he became the first to express and use what is now called Bayes' rule. With it, he updated old knowledge with new, explained phenomena that previous centuries had ascribed to chance or to God's will, and opened the way for future scientific exploration.

Yet Laplace had built his probability theory on intuition. As far as he was concerned, "essentially, the theory of probability is nothing but good common sense reduced to mathematics. It provides an exact appreciation of what sound minds feel with a kind of instinct, frequently without being able to account for it."²⁴ Soon, however, scientists would begin confronting situations that intuition could not easily explain. Nature would prove to be far more complicated than even Laplace had envisioned. No sooner was the old man buried than critics began complaining about Laplace's rule.

many doubts, few defenders

With Laplace gone, Bayes' rule entered a tumultuous period when it was disdained, reformed, grudgingly tolerated, and finally nearly obliterated by battling theorists. Yet through it all the rule chugged sturdily along, helping to resolve practical problems involving the military, communications, social welfare, and medicine in the United States and Europe.

The backdrop to the drama was a set of unsubstantiated but widely circulated charges against Laplace's reputation. The English mathematician Augustus de Morgan wrote in *The Penny Cyclopaedia* of 1839 that Laplace failed to credit the work of others; the accusation was repeated without substantiation for 150 years until a detailed study by Stigler concluded it was groundless. During the 1880s an anti-Napoleonic and antimonarchical Frenchman named Maximilien Marie painted Laplace as a reactionary ultraroyalist; several English and American authors adopted Marie's version unquestioningly. The *Encyclopaedia Britannica* of 1912 asserted that Laplace "aspired to the role of a politician, and . . . degraded to servility for the sake of a riband and a title."¹ In his long-lived but rather fanciful bestseller *Men of Mathematics* the American E. T. Bell titled the chapter on Laplace "From Peasant to Snob . . . Humble as Lincoln, proud as Lucifer." Bell described Laplace as "grandiose," "snobbish," "pompous," "coarse," "smug," "intimate with Napoleon," and "perhaps the most conspicuous refutation of the pedagogical superstition that noble pursuits necessarily ennoble a man's character."² Bell's book, published in 1937, influenced an entire generation of mathematicians and scientists. In the 1960s an Anglo-American statistician Florence Nightingale David wrote without verification that Laplace met with "almost

universal condemnation.”³ A scholarly American biography by historian Charles Coulston Gillispie and two collaborators, Robert Fox and Ivor Grattan-Guinness, hemmed and hawed. It began by stating categorically that “not a single testimonial bespeaking congeniality survives” but ended by listing Laplace’s “close personal attachments with other French scientists,” “warm and tranquil family life,” and the help he gave even to critics of his research.⁴

The realization that Laplace was one of the world’s first modern professional scientists emerged slowly. The statistician Karl Pearson, no shrinking violet, called the author of the *Britannica* article “one of the most superficial writers that ever obscured the history of science. . . . Such statements published by a writer of one nation about one of the most distinguished men of a second nation, and wholly unsubstantiated by references, are in every way deplorable.”⁵ Modern historians have shown that many of the disparaging comments about Laplace’s life and work were false.

Personal insults aside, Laplace launched a craze for statistics that would ultimately inundate both Bayes’ original rule and Laplace’s own version of it. He did so by publicizing in 1827 the then-extraordinary fact that the number of dead letters in the Parisian postal system remained roughly constant from year to year. After the French government published a landmark series of statistics about the Paris region, it appeared that many irrational and godless criminal activities, including thefts, murders, and suicides, were also constants. By 1830 stable statistical ratios were firmly dissociated from divine providence, and Europe was swept by a veritable mania for the objective numbers needed by good government.

Unsettled by rapid urbanization, industrialization, and the rise of a market economy, early Victorians formed private statistical societies to study filth, criminality, and numbers. The chest sizes of Scottish soldiers, the number of Prussian officers killed by kicking horses, the incidence of cholera victims—statistics were easy to collect. Even women could do it. No mathematical analysis was necessary or expected. That most of the government bureaucrats collecting statistics were ignorant of and even hostile to mathematics did not matter. Facts, pure facts, were the order of the day.

Gone was the idea that we can use probability to quantify our lack of knowledge. Gone was the search for causes conducted by Bayes, Price, and Laplace. A correspondent admonished the hospital reformer Florence Nightingale in 1861, “Again I must repeat my objections to intermingling Causation with Statistics. . . . The statistician has nothing to do with causation.”⁶

“Subjective” also became a naughty word. The French Revolution and its

aftermath shattered the idea that all rational people share the same beliefs. The Western world split between Romantics, who rejected science outright, and those who sought certainty in natural science and were enthralled by the objectivity of numbers, whether the number of knifings or of marriages at a particular age.

During the decade after Laplace's death, four European revisionists led the charge against Laplace and probability, the mathematics of uncertainty. John Stuart Mill denounced probability as "an aberration of the intellect" and "ignorance . . . coined into science."⁷ Objectivity became a virtue, subjectivity an insult, and the probability of causes a target of skepticism, if not hostility. Awash in newly collected data, the revisionists preferred to judge the probability of an event according to how frequently it occurred among many observations. Eventually, adherents of this frequency-based probability became known as frequentists or sampling theorists.

To frequentists Laplace was such a towering target that Thomas Bayes' existence barely registered. When critics thought of Bayes' rule, they thought of it as Laplace's rule and focused their criticism on him and his followers. Arguing that probabilities should be measured by objective frequencies of events rather than by subjective degrees of belief, they treated the two approaches as opposites, although Laplace had considered them basically equivalent.

The reformers denounced Laplace's pragmatic simplifications as gross abuses. Two of his most popular applications of probability were condemned wholesale. Laplace had asked: given that the sun has risen thousands of times in the past, will it rise tomorrow? and given that the planets revolve in similar ways around the sun, is there a single cause of the solar system? He did not actually use Bayes' rule for either project, only simple gambling odds. Sometimes, though, he and his followers began answering these questions by assuming 50–50 odds. The simplification would have been defensible had Laplace known nothing about the heavens. But he was the world's leading mathematical astronomer, and he understood better than anyone that sunrises and nebulae were the result of celestial mechanics, not gambling odds. He had also started his study of male and female birthrates with 50–50 odds, although scientists already knew that the likelihood of a male birth is approximately 0.52.

Laplace agreed that reducing scientific questions to chance boosted the odds in favor of his deep conviction that physical phenomena have natural causes rather than religious ones. He warned his readers about it. His followers also sometimes weighted their initial odds heavily in favor of natural

laws and weakened counterexamples. Critics pounded away at the fact that chance was irrelevant to the questions at hand. They identified Bayes' rule with equal priors and damned the entire rule because of them. Few of the critics tried to even imagine other kinds of priors.

Years later John Maynard Keynes studied the complaints made about Laplace's assessment, based on 5,000 years of history, that "it is a bet of 1,825,214 to 1 that [the sun] will rise tomorrow." Summarizing the arguments, Keynes wrote that Laplace's reasoning "has been rejected by [George] Boole on the ground that the hypotheses on which it is based are arbitrary, by [John] Venn on the ground that it does not accord with experience, by [Joseph] Bertrand because it is ridiculous, and doubtless by others also. But it has been very widely accepted—by [Augustus] de Morgan, by [William] Jevons, by [Rudolf] Lotze, by [Emanuel] Czuber, and by Professor [Karl] Pearson—to name some representative writers of successive schools and periods."⁸

Amid the dissension, Laplace's delicate balance between subjective beliefs and objective frequencies collapsed. He had developed two theories of probability and shown that when large numbers are involved they lead to more or less the same results. But if natural science was the route to certain knowledge, how could it be subjective? Soon scientists were treating the two approaches as diametric opposites. Lacking a definitive experiment to decide the controversy and with Laplace demonstrating that both methods often lead to roughly the same result, the tiny world of probability experts would be hard put to settle the argument.

Research into probability mathematics petered out. Within two generations of his death Laplace was remembered largely for astronomy. By 1850 not a single copy of his massive treatise on probability was available in Parisian bookstores. The physicist James Clerk Maxwell learned about probability from Adolphe Quetelet, a Belgian factoid hunter, not from Laplace, and adopted frequency-based methods for statistical mechanics and the kinetic theory of gases. Laplace and Condorcet had expected social scientists to be the biggest users of Bayes' rule, but they were reluctant to adopt any form of probability. An American scientist and philosopher Charles Sanders Peirce promoted frequency-based probability during the late 1870s and early 1880s. In 1891 a Scottish mathematician, George Chrystal, composed an obituary for Laplace's method: "The laws of . . . Inverse Probability being dead, they should be decently buried out of sight, and not embalmed in text-books and examination papers. . . . The indiscretions of great men should be quietly allowed to be forgotten."⁹

For the third time Bayes' rule was left for dead. The first time, Bayes himself had shelved it. The second time, Price revived it briefly before it again died of neglect. This time theoreticians buried it.

The funeral was a trifle premature. Despite Chrystal's condemnation, Bayes' rule was still taught in textbooks and classrooms and used by astronomers because the anti-Bayesian frequentists had not yet produced a systematic, practical substitute. In scattered niches far from the eyes of disapproving theoreticians, Bayes bubbled along, helping real-life practitioners assess evidence, combine every possible form of information, and cope with the gaps and uncertainties in their knowledge.

Into this breach between theoretical disapproval and practical utility marched the French army, under the baton of a politically powerful mathematician named Joseph Louis François Bertrand. Bertrand reformed Bayes for artillery field officers who dealt with a host of uncertainties: the enemy's precise location; air density; wind direction; variations among their hand-forged cannons; and the range, direction, and initial speed of projectiles. In his widely used textbooks Bertrand preached that Laplace's probability of causes was the only valid method for verifying a hypothesis with new observations. He believed, however, that Laplace's followers had lost their way and must stop their practice of indiscriminately using 50–50 odds for prior causes. To illustrate, he told about the foolish peasants of Brittany who, looking for the possible causes of shipwrecks along their rocky coast, assigned equal odds to the tides and to the far more dangerous northwest winds. Bertrand argued that equal prior odds should be confined to those rare cases when hypotheses really and truly were equally likely or when absolutely nothing was known about their likelihoods.

Following Bertrand's strict standards, artillery officers began assigning equal probabilities only for cannons made in the same factory by roughly the same staff using identical ingredients and processes under identical conditions. For the next 60 years, between the 1880s and the Second World War, French and Russian artillery officers fired their weapons according to Bertrand's textbook.

Bertrand's strict Bayesian reforms figured in the Dreyfus affair, the scandal that rocked France between 1894 and 1906. Alfred Dreyfus, a French Jew and army officer, was falsely convicted of spying for Germany and condemned to life imprisonment. Almost the only evidence against Dreyfus was a letter he was accused of having sold to a German military attaché. Alphonse

Bertillon, a police criminologist who had invented an identification system based on body measurements, testified repeatedly that, according to probability mathematics, Dreyfus had most assuredly written the incriminating letter. Bertillon's notions of probability were mathematical gibberish, and he developed ever more fantastical arguments. As conservative antirepublicans, Roman Catholics, and anti-Semites supported Dreyfus's conviction, a campaign to exonerate him was organized by his family, anticlericals, Jews, and left-wing politicians and intellectuals led by the novelist Émile Zola.

At Dreyfus's military trial in 1899, his lawyer called on France's most illustrious mathematician and physicist, Henri Poincaré, who had taught probability at the Sorbonne for more than ten years. Poincaré believed in frequency-based statistics. But when asked whether Bertillon's document was written by Dreyfus or someone else, he invoked Bayes' rule. Poincaré considered it the only sensible way for a court of law to update a prior hypothesis with new evidence, and he regarded the forgery as a typical problem in Bayesian hypothesis testing.

Poincaré provided Dreyfus's lawyer with a short, sarcastic letter, which the lawyer read aloud to the courtroom: Bertillon's "most comprehensible point [is] false. . . . This colossal error renders suspect all that follows. . . . I do not understand why you are worried. I do not know if the accused will be condemned, but if he is, it will be on the basis of other proofs. Such arguments cannot impress unbiased men who have received a solid scientific education."¹⁰ At that point, according to the court stenographer, the courtroom erupted in a "prolonged uproar." Poincaré's testimony devastated the prosecution; all the judges had attended military schools and studied Bayes in Bertrand's textbook.

The judges issued a compromise verdict, again finding Dreyfus guilty but reducing his sentence to five years. The public was outraged, however, and the president of the Republic issued a pardon two weeks later. Dreyfus was promoted and awarded the Legion of Honor, and government reforms were instituted to strictly separate church and state. Many American lawyers, unaware that probability helped to free Dreyfus, have considered his trial an example of mathematics run amok and a reason to limit the use of probability in criminal cases.

As the First World War approached, a French general and proponent of military aviation and tanks, Jean Baptiste Eugène Estienne, developed elaborate Bayesian tables telling field officers how to aim and fire. Estienne

also developed a Bayesian method for testing ammunition. After Germany captured France's industrial base in 1914, ammunition was so scarce that the French could not use wasteful frequency-based methods to test its quality. Mobilized for the national defense, professors of abstract mathematics developed Bayesian testing tables that required destroying only 20 cartridges in each lot of 20,000. Instead of conducting a predetermined number of tests, the army could stop when they were sure about the lot as a whole. During the Second World War, American and British mathematicians discovered similar methods and called them operations research.

Bayes' rule was supposedly still dying on the vine as the First World War approached and the United States faced two emergencies caused by the country's rapid industrialization. In each case self-taught statisticians resorted to Bayes as a tool for making informed decisions, first about telephone communications and second about injured workers.

The first crisis occurred when the financial panic of 1907 threatened the survival of the Bell telephone system owned by American Telephone and Telegraph Company. Alexander Graham Bell's patents had expired a few years earlier, and the company had overexpanded. Only the intervention of a banking consortium led by the House of Morgan prevented Bell's collapse.

At the same time, state regulators were demanding proof of Bell's ability to provide better and cheaper service than local competitors. Unfortunately, Bell telephone circuits were often overloaded in the late morning and early afternoon, when too many customers tried to place calls at the same time. During the rest of the day—80% of the time—Bell's facilities were underutilized. No company could afford to build a system to handle every call that could conceivably be made at peak times.

Edward C. Molina, an engineer in New York City, considered the uncertainties involved. Molina, whose family had emigrated from Portugal via France, was born in New York in 1877. He graduated from a city high school but, with no money for college, got a job first at Western Electric Company and then at AT&T's engineering and research department (later called Bell Laboratories). The Bell System of phone companies was adopting a new mathematical approach to problem solving. Molina's boss, George Ashley Campbell, had studied probability with Poincaré in France but other employees were learning it from the *Encyclopaedia Britannica*. Molina taught himself mathematics and physics and became the nation's leading expert on Bayesian and Laplacean probability.

Unlike many others at the time, he realized that “great confusion exists because many authorities have failed to distinguish clearly between the original Bayes inverse theorem and its subsequent generalization by Laplace. The general theorem embraces, or brings together, both the data obtained from a series of observations and whatever ‘collateral’ information exists in relation to the observed results.”¹¹ As Molina explained, applied statisticians were often forced to make quick decisions based on meager observational data; in such cases, they had to rely on indirect prior knowledge, called collateral information. This could range from assessments of national or historic trends to an executive’s mental health. Methods for utilizing both statistical and nonstatistical types of evidence were needed.

Using Laplace’s formula, Molina combined his prior information about the economics of automating Bell telephone systems with data about telephone call traffic, call length, and waiting time. The result was a cost-effective way for Bell to deal with uncertainties in telephone usage.

Molina then worked on automating Bell’s labor-intensive system. In many cities the company employed 8 to 20% of the female population as telephone operators, switching wires to route calls through trunking facilities to customers in distant exchanges. Operators were in short supply, annual turnover in some cities was 100% or more annually, and wages doubled between 1915 and 1920. Depending on one’s point of view, the work epitomized either opportunities for women or the inhumane pressure of modern technology.

To automate the system Molina conceived of the relay translator, which converted decimally dialed phone numbers into routing instructions. Then he used Bayes to analyze technical information and the economics of various combinations of switches, selectors, and trunking lines at particular exchanges. After women won the right to vote in 1920, Bell feared a backlash if it fired all its operators, so it chose an automating method that merely halved their numbers. Between the world wars, employment of operators dropped from 15 to 7 per 1,000 telephones even as toll call service increased. Probability assumed an important role in the Bell System, and Bayesian methods were used to develop basic sampling theory.

Molina won prestigious awards, but his use of Bayes remained controversial among some Bell mathematicians, and he complained he had trouble publishing his research. Some of his problems may have stemmed from his colorful character. He loved model boats, published articles about Edgar Allan Poe’s use of probability, played the piano expertly, and donated to the Metropolitan Opera in New York. He followed the Russo-Japanese war so

avidly that his colleagues nicknamed him, not fondly, General Molina. When he independently discovered the Poisson distribution, he called it the Molina distribution until he learned to his embarrassment that Laplace's protégé Siméon Denis Poisson had written about it in the 1830s.

Molina's enthusiasm for Bayes-Laplacean probability did not spread to other American corporations. AT&T often regarded his articles about Bayes as proprietary secrets and published them only in in-house publications years after the fact.

While Bayes' rule was helping to save the Bell System, financiers were rushing to build American railroads and industries. Government safety regulations were nonexistent, however, and 1 out of every 318 industrial workers was killed on the job between 1890 and 1910, and many more were injured. The country's labor force suffered more accidents, sickness, invalidity, premature old age, and unemployment than European workers. Yet, unlike most of Europe, the United States had no system for insuring sick and injured workers, and most blue-collar families lived one paycheck away from needing charity. Federal judges ruled that injured employees could sue only if their bosses were personally at fault. In 1898 a U.S. Department of Labor statistician could think of no other social or legal reform in which the United States lagged so far behind other nations.

The tide turned as growing numbers of workers joined the American Federation of Labor and as local juries started awarding generous settlements to their disabled peers. At that point employers decided it was cheaper to treat occupational health as a predictable business expense than to trust juries and encourage unionization. In an avalanche of no-fault laws passed between 1911 and 1920 all but eight states began requiring employers to insure their workers immediately against occupational injuries and illness. This was the first, and for decades the only, social insurance in the United States.

The legislation triggered an emergency. Normally, the price of an insurance premium reflects years of accumulated data about such factors as accident rates, medical costs, wages, industrywide trends, and particulars about individual companies. No such data existed in the United States. Not even the most industrialized states had enough occupational health statistics to price policies for all their industries. The industrial powerhouse of New York State had only enough experience to price policies for machine printers and garment workers; South Carolina had only enough for cotton spinners and weavers; and St. Louis and Milwaukee for beer brewers. In 1909 Nebraska had only

25 small manufacturers of any kind. As an insurance expert wondered, "When will Nebraska be able to determine its pure premium on 'suspenders without buckles,' or Rhode Island on 'butchers' supplies?' And yet rates must be quoted for either, and moreover they must be adequate and equitable."¹²

Data from other areas were seldom relevant. Germany had collected accident statistics for 30 years, but its industrial conditions were safer, and because its data were collected nationwide, premiums could be based on industrywide information. But in the United States, data were collected by state, and Massachusetts's statistics about shoe- and bootmakers were irrelevant to Nevada's metal mines and their high fatality rates because, as one expert reported, "metal mines are so rare in Massachusetts as snakes in Ireland."¹³

Nevertheless, premiums had to be invented—overnight and out of thin air—for almost every sizeable company in the country. It was a nightmare to keep any mathematically trained statistician awake at night—not that the United States had many. Actuaries were often antagonistic to high-level mathematics, and an official complained that accident and fire premiums were typically priced by untrained clerks who used opinion, what they "euphoniously called underwriting judgment,"¹⁴ rather like "woman's intuition . . . ('I don't know why I think so, but I am sure I am right')." ¹⁵ Compounding the crisis, each state legislature mandated its own unique insurance system.

Still, premiums had to be priced accurately: high enough to keep the insurance company solvent for the life of its insureds and individualized enough to reward businesses with good safety records. In an extraordinary feat, Isaac M. Rubinow, a physician and statistician for the American Medical Association, organized by hand the analysis, classification, and tabulation of literally millions of insurance claims, primarily from Europe, as a two- or three-year stopgap until each state could accumulate statistics on its occupational casualties. "Every scrap of information," he said, must be used.¹⁶

Rubinow called together 11 scientifically minded actuaries and formed the Casualty Actuarial Society in 1914. Only seven were college graduates, but their goal was lofty: to set casualty fire and workers' compensation insurance on a sound mathematical basis. Rubinow became the organization's first president but left almost immediately when the insurance industry and the American Medical Association opposed extending social insurance to the sick and aged. Rumors swirled that Rubinow, a Jewish immigrant from Russia, had "socialist tendencies."¹⁷

Albert Wurts Whitney, a specialist in insurance mathematics from Berkeley, replaced Rubinow on a workers' compensation committee. Whit-

ney was an alumnus of Beloit College and had no graduate degrees but had taught mathematics and physics at the universities of Chicago, Nebraska, and Michigan. At the University of California in Berkeley he had taught probability for future insurance professionals. Though not as steeped in the original mathematical literature as Molina at Bell Labs, Whitney was familiar with the theorems of both Laplace and Bayes and knew he should use one of them. He understood something else too. The equations were too complicated for the fledgling workers' compensation movement.

One afternoon in the spring of 1918, during the First World War, Whitney and his committee worked for hours stripping away every possible mathematical complication and substituting dubious simplifications. They agreed to assume that each business in a particular industrial class (for example, all residential roofers) faced equal risks. They would also consider every actuary equally skilled at supplementing injury data with subjective judgments about "non-statistical" or "exogenous material," such as a business owner's drinking habits. This was Bayes' rule where industrywide experience was used as the basis for the prior and the local business's history for the new data. Whitney cautioned, "We know that the [subjective] rates for some classifications are more reliable than for others. [But] it is doubtful whether it is expedient in practice to recognize this fact."¹⁸

By the end of the afternoon the committee had decided to base the price of a customer's premium almost exclusively on the experience of the client's broad classification. Thus a machine shop's premium could be based on data from other, similar businesses or, if it was large enough, its own experience. Combining data from related businesses concentrated the numbers, pulling them closer to the mean and making them more accurate, a subtle "shrinking" effect that Charles Stein would explain in the 1950s. What remained was a stunningly simple formula that a clerk could compute, an underwriter could understand, and a salesman could explain to his customers. The committee proudly named its creation Credibility.

For the next 30 years the first social insurance system in the United States relied on this simplified Bayesian system. In a classic understatement an actuary admitted, "Of course [Credibility's] $Z = P/(P+K)$ is not so great a discovery as $E = mc^2$ nor as unalterably true, but it has made life much easier for insurance men for generations."¹⁹ Some 50 years later statisticians and actuaries would be surprised to discover the Bayesian roots of Credibility.

Next, Whitney worked out methods for weighting each datum as to its subjective believability. Soon actuaries were adventuring "beyond any-

thing that has been proven mathematically. The only demonstration they can make,” an actuary reported later, “is that, in actual practice, it works.”²⁰

Skeptical state officials and insurance underwriters sometimes wondered where these strange Credibility figures came from. One insurance commissioner demanded, “You have supported everything else in the filing with actual experience, where is the experience supporting your Credibility factor?”²¹ Actuaries hastily changed the subject. When Whitney was asked where he got the mathematical principles underlying Credibility, he pointed flipantly to a colleague’s home. “In Michelbacher’s dining room,” he said.

Credibility theory was a practical American response to a uniquely American problem, and it became a cornerstone of casualty and property insurance. As claims accumulated, actuaries could check the accuracy of their premiums by comparing them with actual claims. In 1922 actuaries gained access to an enormous pool of occupational data compiled by the National Council for Compensation Insurance. As the years passed, practicing actuaries had less and less need to understand the relationship between Credibility and Bayes.

While the United States was using Bayes’ theorem for business decisions and France was adapting it for the military, eugenics was shifting Bayes’ story back to its birthplace in Great Britain. There Karl Pearson and Ronald Fisher were developing statistics—the mathematics of uncertainties—into the first information science. Early in the twentieth century, as they created new ways to study biology and heredity, theoreticians would change their attitudes toward Bayes’ rule from tepid toleration to outright hostility.

Karl Pearson (I repeat his first name because his son Egon figures in Bayes’ story too) was a zealous atheist, socialist, feminist, Darwinist, Germanophile, and eugenicist. To save the British Empire, he believed the government should encourage the upper middle class to procreate and the poor to abstain. Karl Pearson ruled Britain’s 30-odd statistical theorists for years. In the process he introduced two generations of applied mathematicians to the kind of feuding and professional bullying generally seen only on middle-school playgrounds.

Contentious, unquenchably ambitious, and craggily determined, Karl Pearson was ambivalent about few things, but Bayes’ rule was one of them. Uniform priors and subjectivity made him nervous. With few other tools available to statisticians, however, he concluded sadly that “the practical man will . . . accept the results of inverse probability of the Bayes-Laplace brand

till better are forthcoming.”²² As Keynes said in *A Treatise on Probability* in 1921, “There is still about it for scientists a smack of astrology, of alchemy.” Four years later the American mathematician Julian L. Coolidge agreed: “We use Bayes’ formula with a sigh, as the only thing available under the circumstances.”²³

Another geneticist, Ronald Aylmer Fisher, eventually contested Karl Pearson’s statistical crown and dealt Bayes’ rule a near-lethal blow. If Bayes’ story were a TV melodrama, it would need a clear-cut villain, and Fisher would probably be the audience’s choice by acclamation.

He didn’t look the part. Even with thick glasses he could barely see three feet and had to be rescued from oncoming buses. His clothes were so rumpled that his family thought he looked like a tramp; he smoked a pipe even while swimming; and if a conversation bored him, he sometimes removed his false teeth and cleaned them in public.

Fisher interpreted any question as a personal attack, and even he recognized that his fiery temper was the bane of his existence. A colleague, William Kruskal, described Fisher’s life as “a sequence of scientific fights, often several at a time, at scientific meetings and in scientific papers.”²⁴ In a basically sympathetic rendering of Fisher’s career, the Bayesian theorist Jimmie Savage said he “sometimes published insults that only a saint could entirely forgive. . . . Fisher burned even more than the rest of us . . . to be original, right, important, famous, and respected. And in enormous measure, he achieved all of that, though never enough to bring him peace.”²⁵ Part of Fisher’s frustration may have arisen from the fact that, on many statistical matters, he was correct.

Fisher was 16 when the family business collapsed. At Cambridge on a scholarship, he became the top mathematics student in his class and, in 1911, the founder and chair of the Cambridge University Eugenics Society. A few years later he solved in one page a problem Karl Pearson had struggled over for years; Pearson thought Fisher’s solution was rubbish and refused to publish it in his prestigious journal *Biometrika*. The two continued to feud as long as they lived. But in straightening out inconsistencies in Karl Pearson’s work, Fisher pioneered the first comprehensive and rigorous theory of statistics and set it on its mathematical and anti-Bayesian course.

The enmity between these two volatile men is striking because both were fervent eugenicists who believed that the careful breeding of British supermen and superwomen would improve the human population and the British Empire. To help support his wife and eight children on a subsistence

farm, Fisher accepted funds from a controversial source, Charles Darwin's son Leonard, who, as honorary president of the Eugenics Education Society, advocated the detention of "inferior types, . . . the sexes being kept apart" to prevent them from bearing children.²⁶ In return for his financial help, Fisher published more than 200 reviews in Darwin's magazine between 1914 and 1934.

In 1919, few jobs were available in statistics or eugenics but Fisher landed a position analyzing fertilizers at Rothamsted Agricultural Experiment Station. Other statistical pioneers worked in breweries, cotton thread and light bulb factories, and the wool industry. Fisher's job was analyzing volumes of data compiled over decades about horse manure, chemical fertilizers, crop rotation, rainfall, temperature, and yields. "Raking over the muck-heap," he called it.²⁷ At first, like Karl Pearson, Fisher used Bayes. But during afternoon teas at Rothamsted soil scientists confronted Fisher with new kinds of literally down-to-earth problems. Fascinated, Fisher worked out better ways to design experiments.

Over the years he pioneered randomization methods, sampling theory, tests of significance, maximum likelihood estimation, analysis of variance, and experimental design methods. Thanks to Fisher, experimental scientists, who had traditionally ignored statistical methods, learned to incorporate them when designing their projects. As the twentieth century's statistical magistrate, Fisher often ended lengthy discussions with a one-word verdict: "Randomize." In 1925 he published a revolutionary manual of new techniques, *Statistical Methods for Research Workers*. A cookbook of ingenious statistical procedures for nonstatisticians, it turned frequency into the de facto statistical method. His first manual sold 20,000 copies, and a second went through seven editions before Fisher's death in 1962. His analysis of variance, which tells how to separate the effects of various treatments, has become one of the natural sciences' most important tools. His significance test and its p -values would be used millions of times even as, over the years, it became increasingly controversial. No one today can discuss statistics—what he called "mathematics applied to observational data"—without using some of Fisher's vocabulary.²⁸ Many of his ideas were solutions to computational problems caused by the limitations of the era's desk calculators. Soon statistical departments rang with the music of bells activated by mechanical calculating machines at every step of their Fisherian calculations.

Fisher himself became a superb geneticist who did mathematical statistics on the side. He filled his house with cats, dogs, and thousands of

mice for cross-breeding experiments; he could document each animal's pedigree for generations. Unlike Bayes, Price, and Laplace, he did not need to supplement inadequate or conflicting observations with hunches or subjective judgments. His experiments produced small data sets and subsets tightly focused to answer a single question with rigorous mathematics. He dealt with few uncertainties or gaps in data and could compare, manipulate, or repeat his experiments as needed. Thanks to the problems he analyzed, Fisher redefined most uncertainties not by their relative probabilities but by their relative frequencies. He brought to fruition Laplace's frequency-based theories, the methods Laplace himself preferred toward the end of his life.

After 15 years at Rothamsted, Fisher moved, first, to University College London and then to Cambridge as professor of genetics. Today, statisticians regard him as one of the great minds of the twentieth century and a "mythical aura" surrounds both him and Karl Pearson.²⁹ The aura around Fisher is a trifle tarnished, though. He left his family on the farm with a precarious, barebones allowance. As a colleague wrote, "If only . . . if only . . . RAF had been a nicer man, if only he had taken pains to be clearer and less enigmatic, if only he had not been obsessed with ambition and personal bitternesses. If only. But then we might not have had Fisher's magnificent achievements."³⁰

Lashing out at Bayes' rule, Fisher called it an "impenetrable jungle" and "a mistake, perhaps the only mistake to which the mathematical world has so deeply committed itself."³¹ Equal priors constituted "staggering falsity."³² "My personal conviction," he declared, is "that the theory of inverse probability is founded upon an error, and must be wholly rejected."³³ A scholarly statistician, Anders Hald, politely lamented "Fisher's arrogant style of writing."³⁴ Although Fisher's work had many Bayesian elements, he battled Bayes for decades and rendered it virtually taboo among respectable statisticians. His constant readiness to quarrel made it hard for opponents to engage him. Bayesians were not alone in concluding that Fisher adopted some of his positions "simply to avoid agreeing with his opponents."³⁵

Driven by the need to deal with uncertainties and save time and money, frequency-based sampling theorists enjoyed a golden age during the 1920s and 1930s. Fisher liberated scientists to summarize and draw conclusions without having to deal with Bayes' messy prior prejudices and hunches. And thanks to his insistence on mathematical rigor, statistics was becoming, if not quite "real mathematics," at least a distinct mathematical discipline, mathematics as applied to data.

The feud between Karl Pearson and Fisher entered its second generation

when Karl's son Egon became another victim of Fisher's wrath. Unlike his father, Egon Pearson was a modest, even self-effacing gentleman. At first, like his father and Fisher earlier in their careers, Egon Pearson frequently used Bayes' rule. In 1925 he published the most extensive exploration of Bayesian methods conducted between Laplace in the 1780s and the 1960s. Using priors for a series of seemingly whimsical experiments, he calculated such probabilities as the fraction of London taxicabs with LX license plates; men smoking pipes on Euston Road; horse-drawn vehicles on Gower Street; chestnut colts born to bay mares; and hounds with fawn-spotted coats. His experiments had a serious purpose, though. He was looking at all sorts of binomial problems, "working backwards" to find "nature's prior," one that anyone with a binomial problem could use. He concluded that more data had to be collected, but no one took up the challenge. Instead, Egon Pearson busied himself trying to make Fisher's work more mathematically rigorous, thereby enraging both Fisher and his father.

Egon Pearson and a Polish mathematician, Jerzy Neyman, teamed up in 1933 to develop the Neyman-Pearson theory of hypothesis testing. Until then, statisticians had tested one hypothesis at a time and either accepted or rejected it without considering alternatives. Egon Pearson's idea was that the only correct reason for rejecting a statistical hypothesis was to accept a more probable one. As he, Neyman, and Fisher developed it, the theory became one of the twentieth century's most influential pieces of applied mathematics. Egon Pearson was afraid of contradicting his father, however. His "fears of K.P. and R.A.F." precipitated a psychological crisis in 1925 and 1926: "I had to go through the painful stage of realizing that K.P. could be wrong . . . and I was torn between conflicting emotions: a. finding it difficult to understand R. A. F., b. hating him for his attacks on my paternal 'god,' c. realizing that in some things at least he was right."³⁶ To placate his father, Egon gave up the woman he loved; they married many years later. And he was so afraid to submit articles to his father's *Biometrika* that he and Neyman published their own journal, *Statistical Research Memoirs*, for two years between 1936 and 1938 and ceased publication only after Karl Pearson died.

Over the years Fisher, Egon Pearson, and Neyman would develop a host of powerful statistical techniques. Fisher and Neyman became fervent anti-Bayesians who limited themselves to events that could theoretically be repeated many times; regarded samples as their only source of information; and viewed each new set of data as a separate problem, to be used if the data were powerful enough to provide statistically significant conclusions and

discarded if not. As anti-Bayesians, they banned subjective priors, although they did not argue with Bayes' theorem when the priors were known; the difficulties and controversies arose when the prior probabilities were unknown. Neyman, for example, denounced Bayes' equal-prior shortcut as "illegitimate."³⁷

In addition, in a deep philosophical divide between the two methods, frequentists asked for the probability of a data set given full knowledge of its probable causes, while Bayesians could ask for better knowledge of the causes in light of the data. Bayesians could also consider the probability of a single event, like rain tomorrow; encapsulate subjective information in priors; update their hunches with new information; and include every datum possible because each one might change the answer by a small amount.

In time, however, Fisher and Neyman also split asunder, starting yet another juicy 30-year feud. Their views on testing, which could be an order of magnitude apart, formed the crux of their bitter fight. According to Neyman, though, the argument began because Fisher demanded that Neyman lecture only from Fisher's book. When Neyman refused, Fisher promised to oppose him "in all my capacities."

In an argument during a meeting of the Royal Society on March 28, 1934, a secretary took the customary word-for-word notes for verbatim publication. Neyman presented a paper arguing that the Latin square (a technique invented by Fisher for experimental design) was biased. Fisher immediately marched to a blackboard, drew a Latin square and, using a simple argument, showed that Neyman was wrong. But Fisher was far from polite. He complained sarcastically that "he had hoped that Dr. Neyman's paper would be on a subject with which the author was fully acquainted, and on which he could speak with authority. . . . Dr. Neyman had been somewhat unwise in his choice of topics." Fisher couldn't seem to stop. He kept on: "Dr. Neyman had arrived or thought he had arrived at . . . Apart from its theoretical defects . . . the apparent inability to grasp the very simple argument . . . How had Dr. Neyman been led by his symbolism to deceive himself on such a simple question?" and on and on.³⁸

By 1936 the feud between the Neyman camp and the Fisherites was an academic cause célèbre. The two groups occupied different floors of the same building at University College London but they never mixed. Neyman's group met in the common room for India tea between 3:30 and 4:15 p.m. Fisher's group sipped China tea from then on. They were fighting over scraps. The statistics building had no potable water, so little electricity that blackboards

were unreadable after dark, and so little heat in winter that overcoats were worn inside.

George Box, who straddled both groups (he studied under Egon Pearson, became a Bayesian, and married one of Fisher's daughters), said that Fisher and Neyman "could both be very nasty and very generous at times." Because Neyman was decision-oriented and Fisher was more interested in scientific inference, their methodologies and types of applications were different. Each was doing what was best for the problems he was working on, but neither side made any attempt to understand what the other was doing. A popular in-house riddle described the situation: "What's the collective noun for a group of statisticians?" "A quarrel."³⁹

Shortly before the Second World War, Neyman moved to the University of California at Berkeley and transformed it into an anti-Bayesian powerhouse. The Neyman-Pearson theory of tests became the Berkeley school's glory and emblem. The joke, of course, was that the University of Berkeley's namesake, the Bishop Berkeley, had disapproved of calculus and mathematicians.

The golden age of probability theory had turned into a double-fronted attack by two camps of warring frequentists united in their abhorrence of Bayes. In the maelstrom, the lack of reasoned discourse among the leaders of statistical mathematics delayed the development of Bayes' rule for decades. Caught in the infighting, the rule was left to find its way alone, stymied and disparaged.

Yet even as the frequentists' assault laid it low, the first glimmerings of a revival flickered here and there, quietly and almost unnoticed. In a remarkable confluence of thinking, three men in three different countries independently came up with the same idea about Bayes: knowledge is indeed highly subjective, but we can quantify it with a bet. The amount we wager shows how much we believe in something.

In 1924 a French mathematician, Émile Borel, concluded that a person's subjective degree of belief could be measured by the amount he was willing to bet. Borel argued that applying probability to real problems, such as insurance, biology, agriculture, and physics, was far more important than mathematical theorizing. He believed in rational behavior and lived as he taught. At the height of a scandal over Marie Curie's possible affair with another scientist, Borel sheltered her and her daughters; in reaction, the minister of public instruction threatened to fire him from his teaching post at the *École Normale Supérieure*, a leading school of mathematics and science.⁴⁰

Between the two world wars, Borel was a member of the French Chamber of Deputies and a minister of the navy and helped direct national policy toward research and education. Imprisoned briefly during the Second World War by the pro-Nazi Vichy government, he later received the Resistance Medal.

Two years after Borel's suggestion a young English mathematician and philosopher named Frank P. Ramsey made the same suggestion. Before he died at the age of 26 following surgeries for jaundice, he wondered how we should make decisions in the face of uncertainty. In an informal talk to students at the Moral Sciences Club at Cambridge University in 1926 Ramsey suggested that probability was based on personal beliefs that could be quantified by a wager. Such extreme subjectivity broke radically with previous thinkers like Mill, who had denounced subjective probabilities as the abhorrent quantification of ignorance.

Ramsey, who in his brief career also contributed to economics, logic, and philosophy, believed uncertainty had to be described in terms of probability rather than by tests and procedures. By talking about a measure of belief as a basis for action and by introducing a utility function and the maximization of expected utility, he showed how to act in the face of uncertainty. Neither Bayes nor Laplace had ventured into the world of decisions and behavior. Because Ramsey worked in Cambridge, England, the history of Bayes' rule might have been quite different had he lived longer.

At almost the same time as Borel and Ramsey, an Italian actuary and mathematics professor, Bruno de Finetti, also suggested that subjective beliefs could be quantified at the racetrack. He called it "the art of guessing."⁴¹ De Finetti had to deliver his first important paper in Paris because the most powerful Italian statistician, Corrado Gini, regarded his ideas as unsound. (In Gini's defense, de Finetti told colleagues he was convinced Gini had "the evil eye.")⁴² De Finetti, considered the finest Italian mathematician of the twentieth century, wrote about financial economics and is credited with putting Bayes' subjectivity on a firm mathematical foundation.

Not even probability experts, however, noticed these outbursts of subjective betting. During the 1920s and 1930s the anti-Bayesian trio of Fisher, Egon Pearson, and Neyman attracted all the attention. Ramsey, Borel, and de Finetti worked outside English-speaking statistical circles.

Another outsider carved out a safe haven for Bayes in paternity law, a small and inconspicuous corner of the American judicial system. Paternity law asks, Did this man father this child? And if so, how much should he pay in child support? In 1938 a Swedish professor of genetics and psychiatry

named Erik Essen-Möller developed an index of probability that was mathematically equivalent to Bayes' theorem. For 50 years, until DNA profiling became available, American lawyers used the Essen-Möller index without knowing its Bayesian paternity. In the U.S. Uniform Parentage Act, Bayes even became a model for state legislation. Because paternity lawyers began by assigning 50–50 odds to the man's innocence, the index favored fathers even though Essen-Möller believed that "mothers more frequently accuse true than false fathers."⁴³ Bayesian paternity law was also used in immigration and inheritance cases and in cases where a child was born as a result of rape. Today, DNA evidence typically gives paternity probabilities of 0.999 or more.

Yet another outsider, Lowell J. Reed, a medical researcher at Johns Hopkins University in Baltimore, dramatized the shortcomings of frequentism and the value of Bayes in 1936. Reed, a member of the department of biostatistics, wanted to determine the X-ray dosages that would kill cancerous tumors but leave patients unharmed. He had no precise exposure records, however, and the effects of low doses were not understood. Reed normally used frequency methods and repeated tests on fruit flies, protozoa, and bacteria; but to ascertain doses for humans he would have to use expensive mammals. With Bayes, he determined the most therapeutic doses for human cancer patients by sacrificing a relatively small number of cats, 27. But Reed worked outside the statistical mainstream, used Bayes only occasionally, and had little influence on statistics. Even Ramsey, Borel, de Finetti, and Essen-Möller had to wait decades before the importance of their work was recognized.

It was a geophysicist, Harold Jeffreys, who almost singlehandedly kept Bayes alive during the anti-Bayesian onslaught of the 1930s and 1940s. Cambridge University students liked to joke that they had the world's two greatest statisticians, although one was a professor of astronomy and the other a professor of genetics. Fisher was the geneticist. Jeffreys was an Earth scientist who studied earthquakes, tsunamis, and tides. He said he qualified for the astronomy department "because Earth is a planet."⁴⁴

Thanks in large part to Jeffreys's quiet, gentlemanly personality he and Fisher became friends even though they disagreed, irrevocably and vociferously, over Bayes. Jeffreys said he told Fisher that "on most things we should agree and when we disagreed, we would both be doubtful. After that, Fisher and I were great friends."⁴⁵ For example, Jeffreys believed that Fisher's maximum likelihood method was basically Bayesian, and he often used it because with large samples the prior did not matter and the two techniques produced approximately the same results. They differed, however, when

small amounts of data were involved. Years later others would dramatize situations where Jeffreys's and Fisher's significance test results can differ by an order of magnitude.

Aside from their views on Bayes, Jeffreys and Fisher had much in common. Both were practicing scientists who manipulated statistical data; neither was a mathematician or statistician. Both were educated at Cambridge; Jeffreys, in fact, never left and was a fellow there for 75 years, longer than any other professor. Neither man was outgoing; both were appalling lecturers, their feeble voices inaudible beyond a few rows, and a student once counted Jeffreys mumbling "er" 71 times in five minutes. Both were knighted for their work.

Of the two, Jeffreys led the richer personal life. At the age of 49 he married his longtime collaborator, the mathematician Bertha Swirles; they proofread their monumental *Methods of Mathematical Physics* during all-night stints as air raid wardens during the Second World War. He enjoyed annotating discrepancies in detective novels, singing tenor in choral societies, botanizing, walking, traveling, and, until he was 91, bicycling to work.

Like Laplace, Jeffreys studied the formation of the Earth and planets in order to understand the origin of the solar system. He became involved in statistics because he was interested in how earthquake waves travel through the Earth. A major earthquake generates seismic waves that can be recorded thousands of miles away. By measuring their arrival times at different stations, Jeffreys could work back to determine the earthquake's likely epicenter and the likely composition of the Earth. It was a classic problem in the inverse probability of causes. In 1926 Jeffreys inferred that Earth's central core is liquid—probably molten iron, probably mixed with traces of nickel.

As one historian said, "Perhaps in no other field were as many remarkable inferences drawn from so ambiguous and indirect data."⁴⁶ Signals were often difficult to interpret, and seismograph machines differed greatly. Earthquakes, which often occurred far apart under very different conditions, were hardly repeatable. Jeffreys's conclusions involved far more uncertainty than Fisher's breeding experiments, which were designed to answer precise, repeatable questions. Like Laplace, Jeffreys spent a lifetime updating his observations with new results. He wrote, "The propositions that are in doubt . . . constitute the most interesting part of science; every scientific advance involves a transition from complete ignorance, through a stage of partial knowledge based on evidence becoming gradually more conclusive, to the stage of practical certainty."⁴⁷

Working on his office floor ankle-deep with papers, Jeffreys composed *The Earth: Its Origin, History, and Physical Constitution*, the standard work on the planet's structure until plate tectonics was discovered in the 1960s. (Sadly, while Jeffreys played the hero defending Bayes, he opposed the idea of continental drift as late as 1970, when he was 78, because he thought it meant the continents would have to push their way through viscous liquid.)

While analyzing earthquakes and tsunamis, Jeffreys worked out a new, objective form of Bayes for scientific applications and devised formal rules for selecting priors. As he put it, "Instead of trying to see whether there was any more satisfactory form of the prior probability, a succession of authors have said that the prior probability is nonsense and therefore that the principle of inverse probability, which cannot work without [priors], is nonsense too."⁴⁸

Jeffreys considered probability appropriate for all uncertainty, even something as apparently certain as a scientific law, whereas frequentists usually restricted probability to the uncertainties associated with theoretically repeatable data. As the statistician Dennis Lindley wrote, Jeffreys "would admit a probability for the existence of the greenhouse effect, whereas most [frequentist] statisticians would not and would confine their probabilities to the data on CO₂, ozone, heights of the oceans, etc."⁴⁹

Jeffreys was particularly annoyed by Fisher's measures of uncertainty, his "p-values" and significance levels. The p-value was a probability statement about data, given the hypothesis under consideration. Fisher had developed them for dealing with masses of agricultural data; he needed some way to determine which should be trashed, which filed away, and which followed up on immediately. Comparing two hypotheses, he could reject the chaff and save the wheat.

Technically, p-values let laboratory workers state that their experimental outcome offered statistically significant evidence against a hypothesis if the outcome (or a more extreme outcome) had only a small probability (under the hypothesis) of having occurred by chance alone.

Jeffreys thought it very strange that a frequentist considered possible outcomes that had not occurred. He wanted to know the probability of his hypothesis about the epicenter of a particular earthquake, given his information about the arrival times of tsunamis caused by the earthquake. Why should possible outcomes that had not occurred make anyone reject a hypothesis? Few researchers repeated—or could repeat—an experiment at random many, many more times. "Imaginary repetitions," a critic called them. Bayesians considered data as fixed evidence, not as something that can

vary. Jeffreys certainly could not repeat a particular earthquake. Moreover, the p -value is a statement about data, whereas Jeffreys wanted to know about his hypothesis given his data. As a result, Jeffreys proposed using only observed data with Bayes' rule to compute the probability that the hypothesis was true.

Newton, as Jeffreys pointed out, derived his law of gravity 100 years before Laplace proved it by discovering Jupiter's and Saturn's 877-year cycle: "There has not been a single date in the history of the law of gravitation when a modern significance test would not have rejected all laws [about gravitation] and left us with no law."⁵⁰

Bayes, on the other hand, "makes it possible to modify a law that has stood criticism for centuries without the need to suppose that its originator and his followers were useless blunderers."⁵¹

Jeffreys concluded that p -values fundamentally distorted science. Frequentists, he complained, "appear to regard observations as a basis for possibly rejecting hypotheses, but in no case for supporting them."⁵² But odds are that at least some of the hypotheses Fisher rejected were worth investigating or were actually true.

A frequentist who tests a precise hypothesis and obtains a p -value of .04, for example, can consider that significant evidence against the hypothesis. But Bayesians say that even with a .01 p -value (which many frequentists would see as extremely strong evidence against a hypothesis) the odds in its favor are still 1 to 9 or 10—"not earth-shaking," says Jim Berger, a Bayesian theorist at Duke University. P -values still irritate Bayesians. Steven N. Goodman, a distinguished Bayesian biostatistician at Johns Hopkins Medical School, complained in 1999, "The p -value is almost nothing sensible you can think of. I tell students to give up trying."⁵³

Jeffreys was making Laplace's probability of causes useful for practicing scientists, even as Fisher was doing the same for Laplace's frequency-based methods. The difference was that Fisher used the word "Bayes" as an insult, while Jeffreys called it the Pythagorean theorem of probability theory. As the first since Laplace to apply formal Bayesian theory to a variety of important scientific problems, Jeffreys became the founder of modern Bayesian statistics.

Statistically, the lines were drawn. Jeffreys and Fisher, two otherwise cordial Cambridge professors, embarked on a two-year debate in the Royal Society's *Proceedings*. Jeffreys may have been shy and uncommunicative, but when he was sure of himself he dug in, placidly but implacably. Fisher remained his usual "volcanic and paranoid" self.⁵⁴ Both men were magnifi-

cent scientists, the world's leading statisticians, and each used the methods best suited to his respective field. Yet neither could see the other's point of view. Like gladiators of old, they hurled impassioned articles back and forth, criticizing one another and issuing formal rejoinders, probing rebuttals, and brilliant clarifications—until Royal Society editors threw up their hands in exasperation and ordered the warriors to cease and desist.

After the grand debate, Jeffreys wrote a monumental book, *Theory of Probability*, which remained for years the only systematic explanation of how to apply Bayes to scientific problems. Fisher complained publicly that Jeffreys makes “a logical mistake on the first page which invalidates all the 395 formulae in his book.”⁵⁵ The mistake, of course, was to use Bayes' theorem. Summarizing Jeffreys's books, Lindley said, “De Finetti is a master of theory, Fisher a master of practice, but Jeffreys is brilliant at both.”⁵⁶

The Fisher–Jeffreys debate ended inconclusively. Practically speaking, however, Jeffreys lost. For the next decade and for a variety of reasons frequentism almost totally eclipsed Bayes and the inverse probability of causes.

First, Fisher was persuasive in public, while the mild-mannered Jeffreys was not: people joked that Fisher could win an argument even when Jeffreys was right. Another factor was that social scientists and statisticians needed objective methods in order to establish themselves as academically credible in the 1930s. More particularly, physicists developing quantum mechanics were using frequencies in their experimental data to determine the most probable locations of electron clouds in nuclei. Quantum mechanics was new and chic, and Bayes was not.

In addition, Fisher's techniques, written in a popular style with minimal mathematics, were easier to apply than those of Jeffreys. A biologist or psychologist could easily use Fisher's manual to determine whether results were statistically significant. To use Jeffreys's rather opaque and mathematical approach, a scientist had to choose among five nuanced categories: the evidence against the hypothesis is “not worth more than a bare mention” or it is substantial, strong, very strong, or decisive.⁵⁷ Characteristically, Jeffreys tucked the five categories into appendix B of his book.

Finally and most important, Jeffreys was interested in making inferences from scientific evidence, not in using statistics to guide future action. To him, decision making—so important for the rise of mathematical statistics during the Second World War and the Cold War—was irrelevant. Others parted at the same divide: a big reason for the feud between Fisher and Neyman, for example, was decision theory.

All these factors left Jeffreys almost totally isolated from statistical theorists. His link with Fisher was their interest in applying statistics to science. Jeffreys knew Ramsey and visited him as he was dying in the hospital, but neither realized the other was working on probability theory; in any case, Jeffreys was interested in scientific inference and Ramsey in decision making. Jeffreys and de Finetti worked on similar probability issues during the 1930s, but Jeffreys did not even know the Italian's name for half a century and would have rejected outright de Finetti's subjectivity. Most statisticians ignored Jeffreys's book on probability theory for years; he said that "they were completely satisfied with frequency theories."⁵⁸ Jeffreys accepted a medal from the Royal Statistical Society, but attended none of its meetings. Geophysicists did not know about his probability work; a surprised geologist once asked Lindley, "You mean that your Jeffreys is the same Jeffreys as mine?"⁵⁹

By 1930 Jeffreys was truly a voice in the wilderness. Most statisticians were using the powerful body of ideas developed by the anti-Bayesian trio. Jeffreys's great book *Theory of Probability* was published as part of a series of physics books, not statistics books. It also appeared in the last year of peace, just before the start of the Second World War and a new opportunity for Bayes' rule.